

Diffusione delle Disinformazioni

Stefano Mizzaro

DMIF – UNIUD

mizzaro@uniud.it

Social Computing – Lezione 15

21 Novembre 2025

Calendario lezioni

«È difficile fare
previsioni, soprattutto
sul futuro» [Bohr ?]

- 15. Ven 21/11 (oggi)
 - Fine Parte 1
- 16. Lun 24/11 ["Lab.", Hafsa]
 - (Richiami Python, Pandas)
 - NetworkX
- 17. Ven 28/11 ["Lab.", Hafsa]
 - Pyviz
 - Progetto
- 18. Lun 1/12
 - Inizio Parte 2



Portate i vostri laptop!
(per esercizi, ecc.)

Riassunto

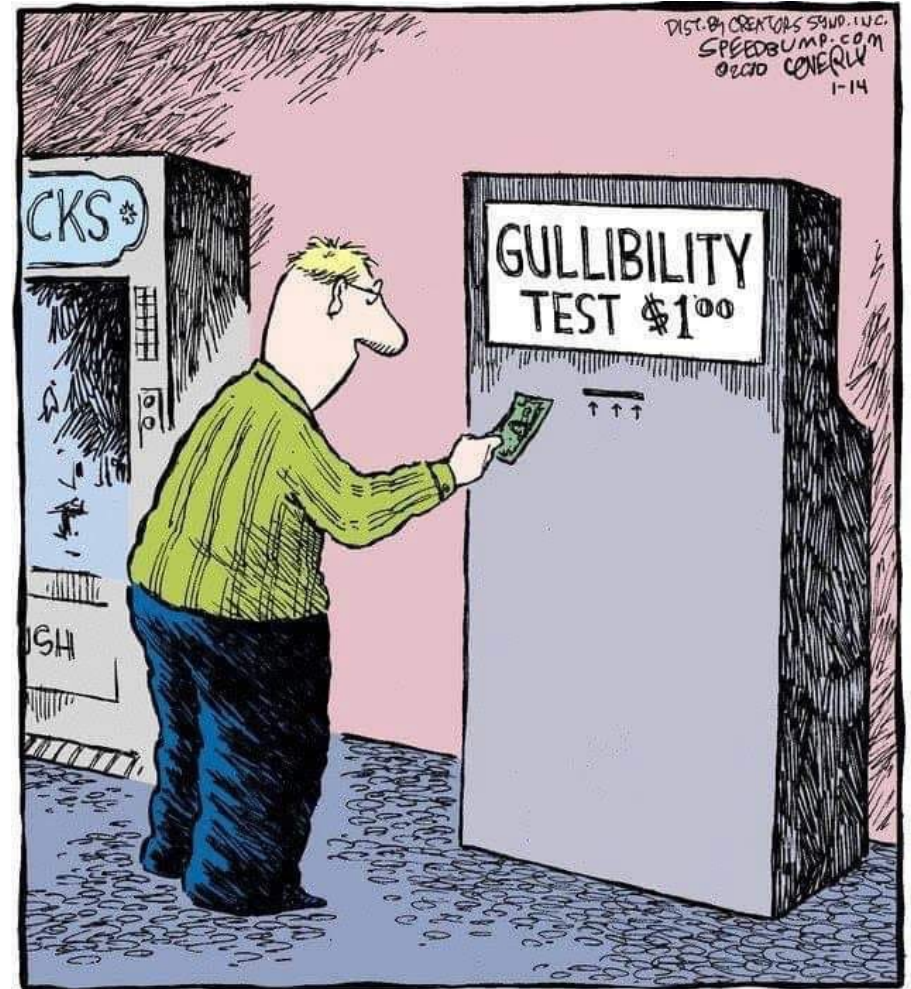
- 1. Intro a Social Computing
- 2. Intro a Social Media
 - Termini, Definizioni, Esempi, Classificazioni, Storia, Statistiche
- 3. Grafi e Reti
 - Definizioni di base, Rete = Grande grafo nel mondo reale
- 4, 5, 6. Misure delle reti
 - Centralità, transitività e reciprocità, C, distanza, degree distribution, ...
- 7, 8, 9. Modelli delle reti
 - Regolari, casuali, piccolo mondo, scale-free/power-law
- 10. Comunità. Il grafo di Facebook
- 11, 12, 13, 14. Diffusione informazioni
 - Herd behavior, Information cascade (ICM), Epidemie, Contagio sociale, Benefici diretti

Questionario

- <https://forms.gle/SfPFyBq84v9gNcudA>

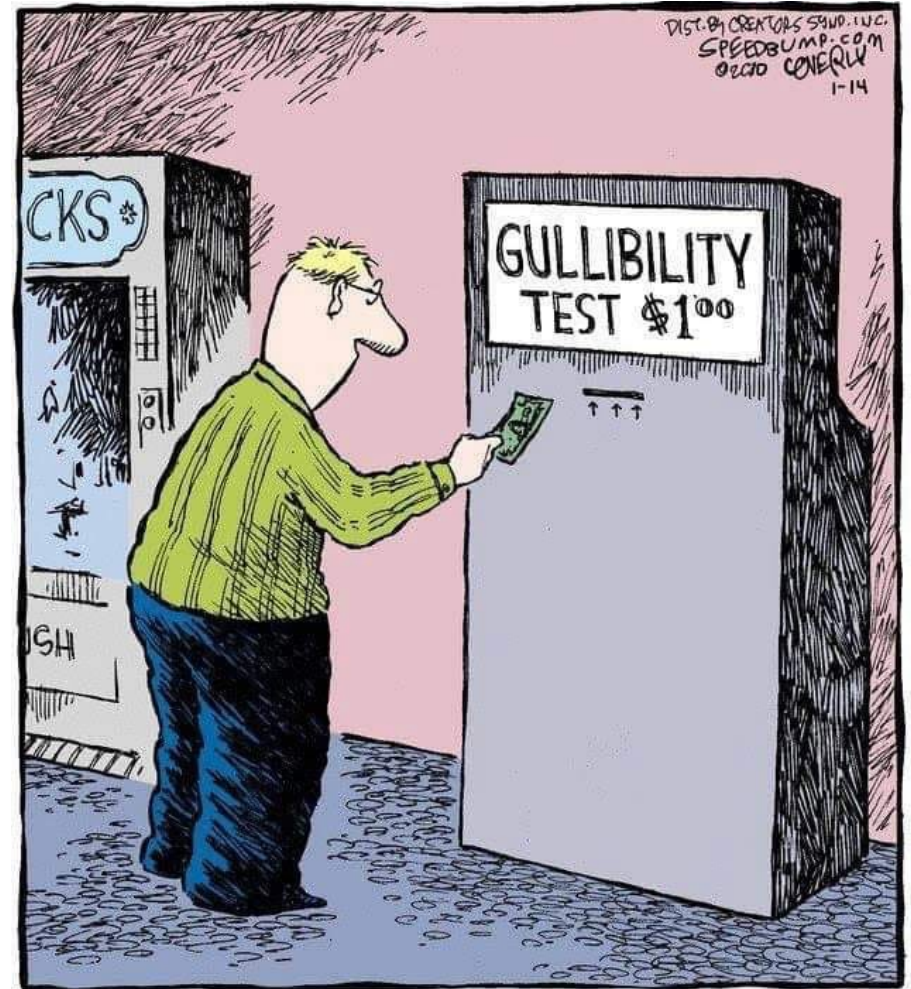
Scaletta

- Intro
 - Terminologia/Definizioni
 - Motivazioni
- Modelli epidemici
- 3 studi sperimentali



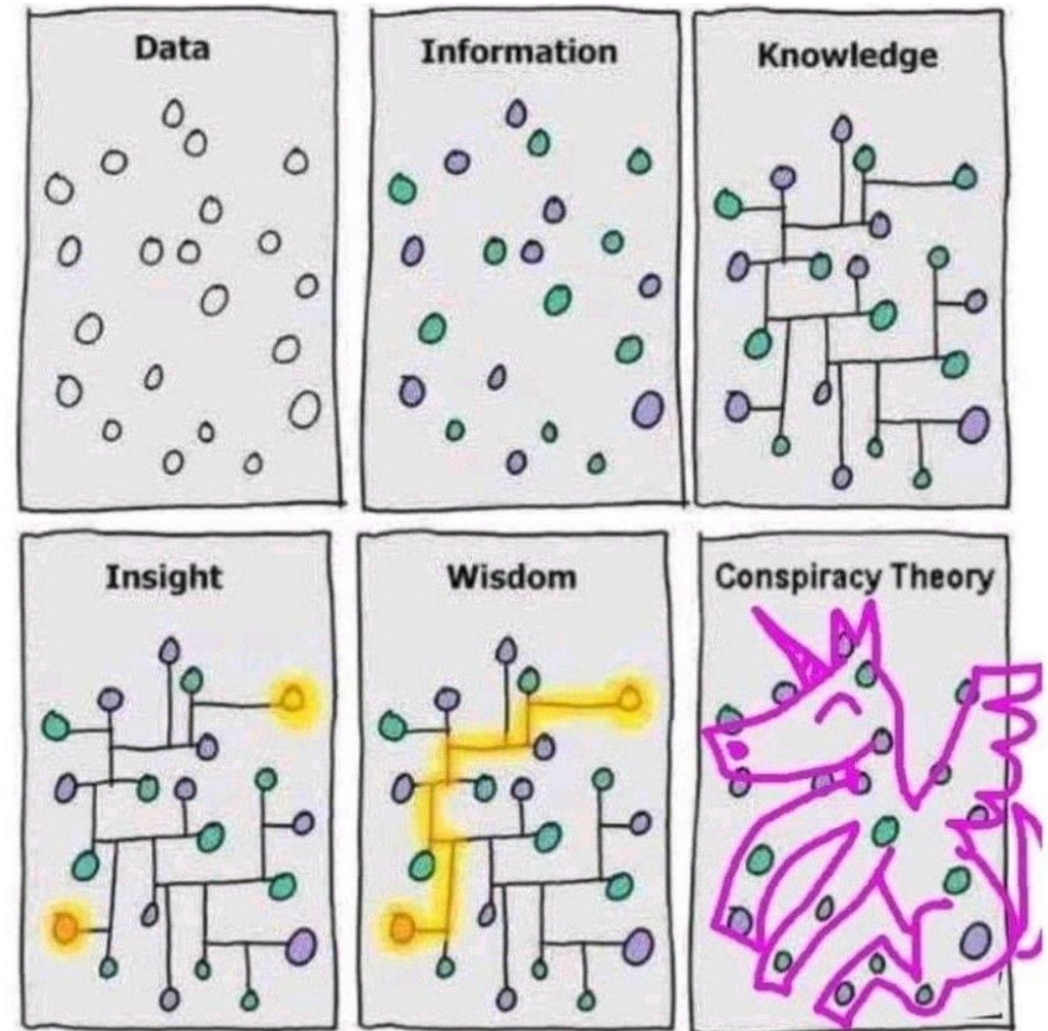
Scaletta

- Intro
 - Terminologia/Definizioni
 - Motivazioni
- Modelli epidemici
- 3 studi sperimentali



Terminologia

- Misinformation
- Disinformation
 - "Fake news"
- ...



Credits



The University Of Sheffield.



WeVerify

**Understanding Online Misinformation:
Major Challenges Ahead**

Carolina Scarton

ROME, Paris, 25/07/2019



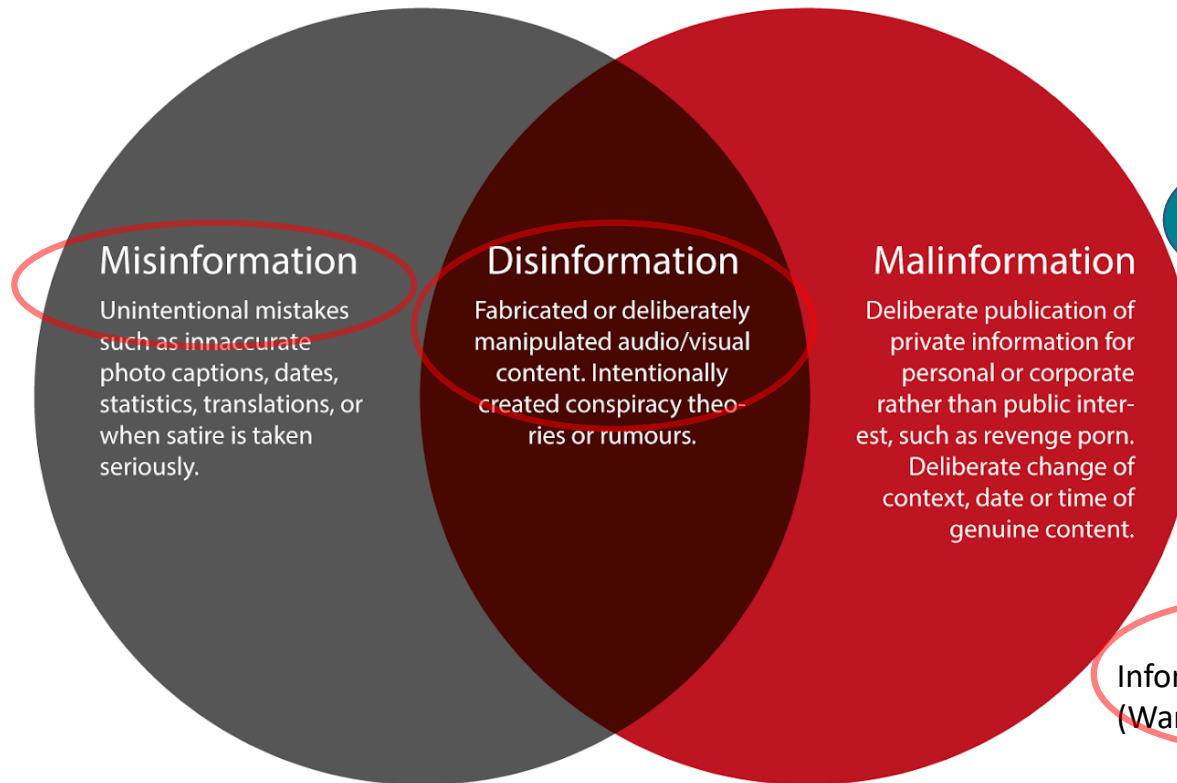
WeVerify project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 825297

TYPES OF INFORMATION DISORDER



FALSENESS

INTENT TO HARM



Very difficult to distinguish
→ mainly between MIS-
and DISinformation

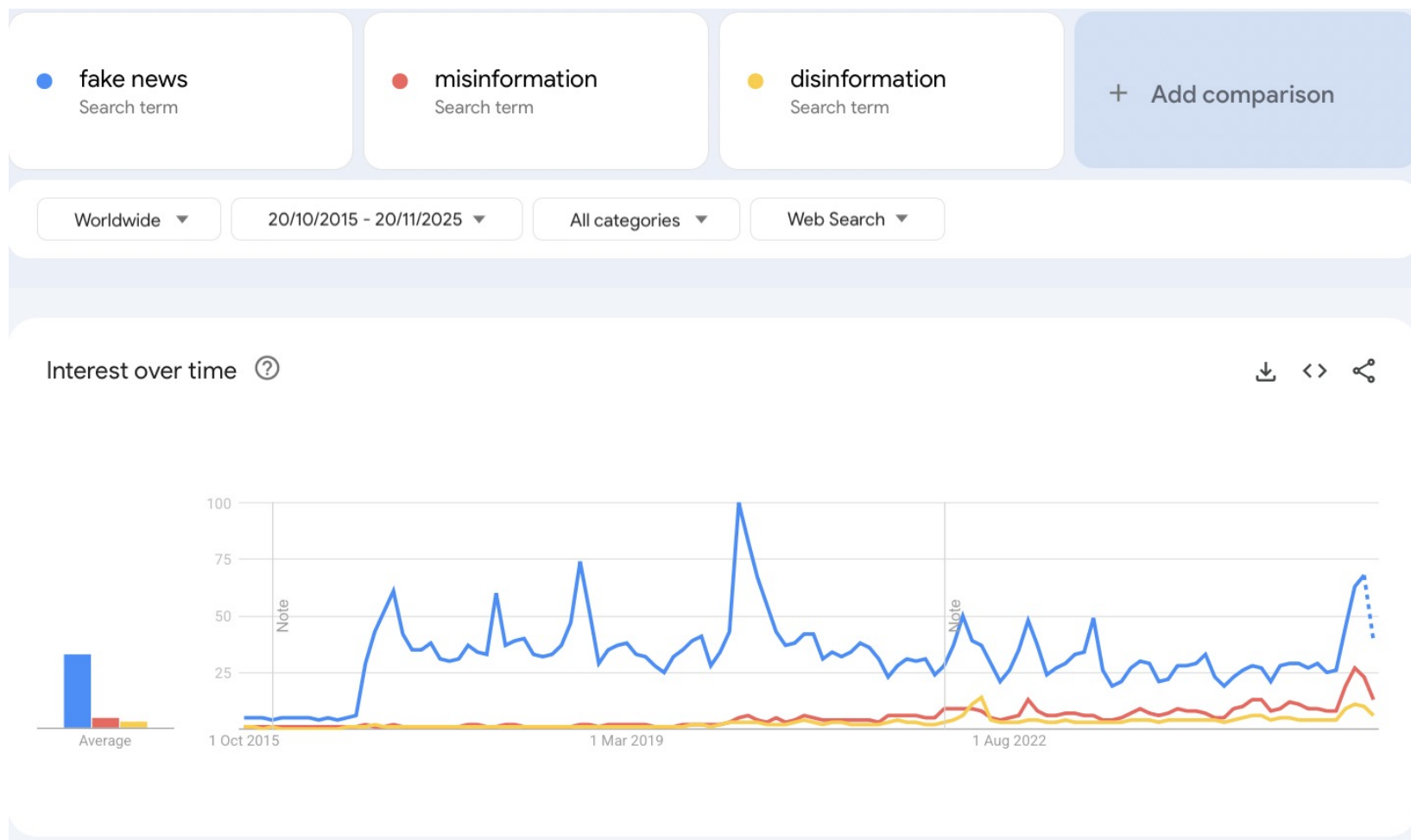
Information disorder theoretical framework
(Wardle, 2017; Wardle & Derakshan, 2017)



Terminologia

- Misinformation, Misinformazione
- **Disinformation, Disinformazione**
 - (quello che vediamo)
- "Fake news" (troppo specifico, e connotato politicamente)
- ...ma alcuni diffondono in buona fede notizie create per essere false → al confine fra Disinformation e Misinformation...

Poi, in pratica...



A paper

- Kai Kupferschmidt, A Field's Dilemmas, *Science*, Vol. 386 Issue 6721, 1 November 2024
- <https://www.science.org/doi/epdf/10.1126/science.adu2100>
- 1st "dilemma": basic definition of misinformation

Definition(s)?!

- Clearly false news masquerading as the real thing?
 - Clearly false news are very few (maybe $< 1\%$)
 - Too restrictive
- Any information that counters the consensus in science?
 - What is consensus?
 - Consensus changes...

Definition(s)?!

- Any information that is false?
 - Too broad (too many things would be called misinformation)
 - But also restrictive again!
 - A factual story (a true story) with misleading headlines (→ex.)
 - This would not be "false"!
 - "There's plenty of examples, where things are true, but they are completely misleading, which is a form of misinformation as well"
 - Clickbait headlines
- ⇒ Falsehood is not a good criterion
 - Or rather not the only one

Example

- In January 2021, the South Florida Sun Sentinel published a story about Gregory Michael, a doctor who had received a shot of Pfizer's COVID-19 vaccine just before the end of 2020 and then died suddenly on 3 January 2021.
- The story was republished by the Chicago Tribune under the headline "A 'healthy' doctor died two weeks after getting a COVID-19 vaccine; CDC is investigating why" and quickly spread online.
- On Facebook it was one of the most shared articles of that year, seen by more than 54 million people in the U.S.
- (The problem was the headline)

Definition(s)?!

- Misinformation as individual, atomic units of information, such as an article or a tweet?
- That overlooks the bigger picture of disinformation:
 - misinformation that is spread deliberately to mislead others
- Disinformation campaigns often work by selectively amplifying certain pieces of news, all or most of which can be true
- "Disinformation is not a piece of content. It's a strategy." [Kate Starbird]

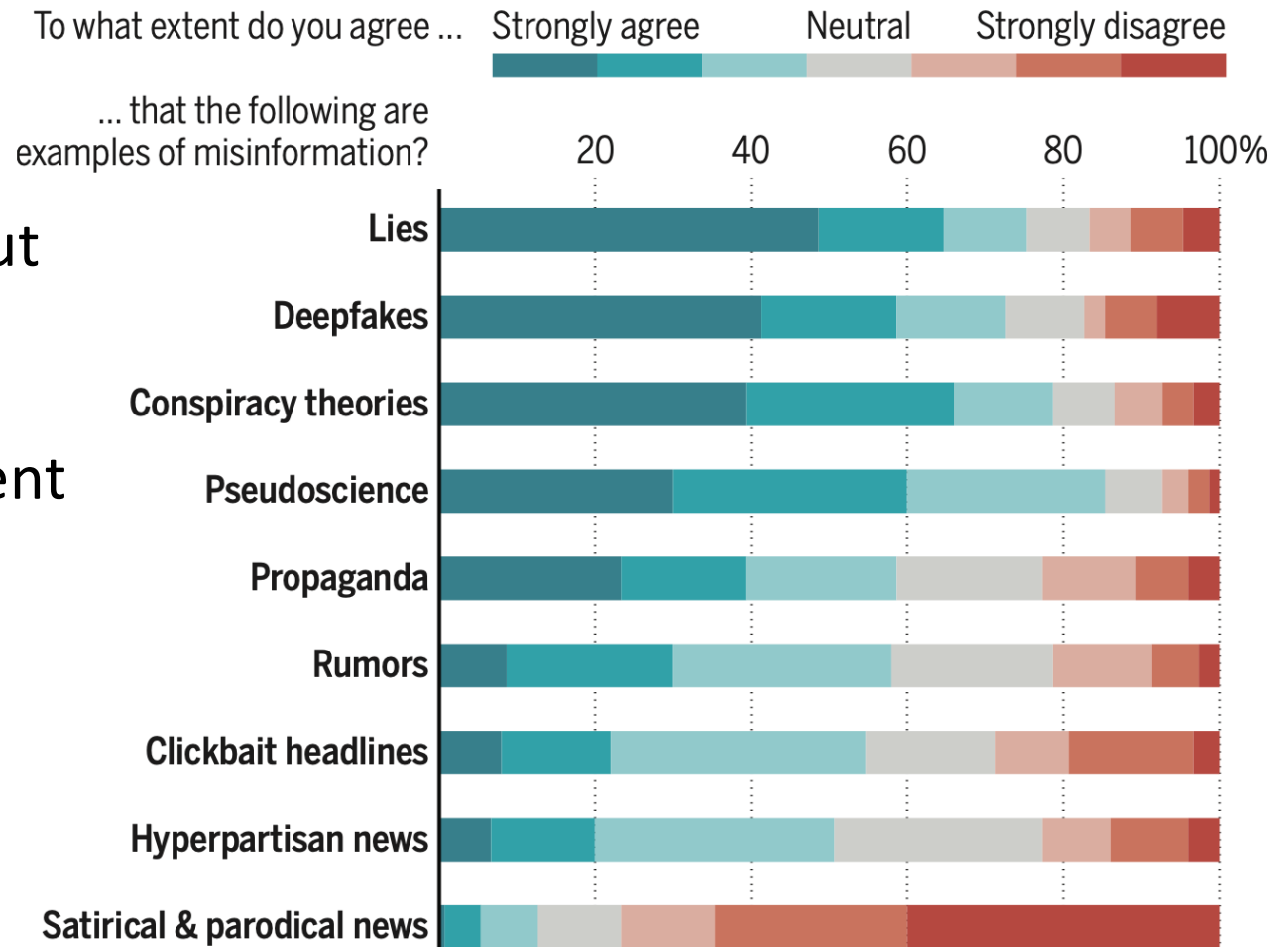
A survey

- Two comments

- 1. There is disagreement about what is misinformation
- 2. There are different "kinds" of misinformation

Defining misinformation

A 2023 survey of 150 misinformation experts found there is often no consensus about which kinds of content count as misinformation.



<https://www.science.org/doi/epdf/10.1126/science.adu2100>

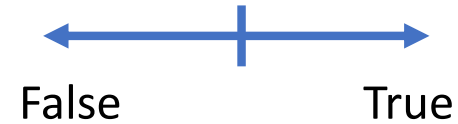
Also: Is truth binary?

- TRUE: True.
- FALSE: False.
- Seems natural
- Mathematical logic



Why two levels?

- True
- Neither true nor false
- False



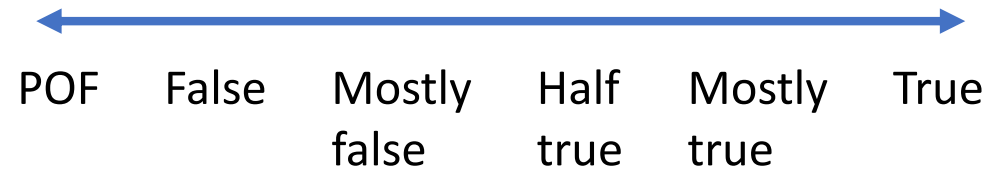
Why two levels?

- Completely True
- Partially True
- Partially False
- Completely False



Truthfulness levels IRL

- TRUE: The statement is accurate and there's nothing significant missing
- MOSTLY TRUE: The statement is accurate but needs clarification or additional information
- HALF TRUE: The statement is partially accurate but leaves out important details or takes things out of context
- MOSTLY FALSE: The statement contains an element of truth but ignores critical facts that would give a different impression
- FALSE: The statement is not accurate
- PANTS ON FIRE: The statement is not accurate and makes a ridiculous claim



Politifact scale

<https://www.politifact.com/>

"Ordinal categories"

6 livelli, e con quelle etichette...

- Politifact: <https://www.politifact.com>
- IRL, non solo curiosità accademica



POLITIFACT

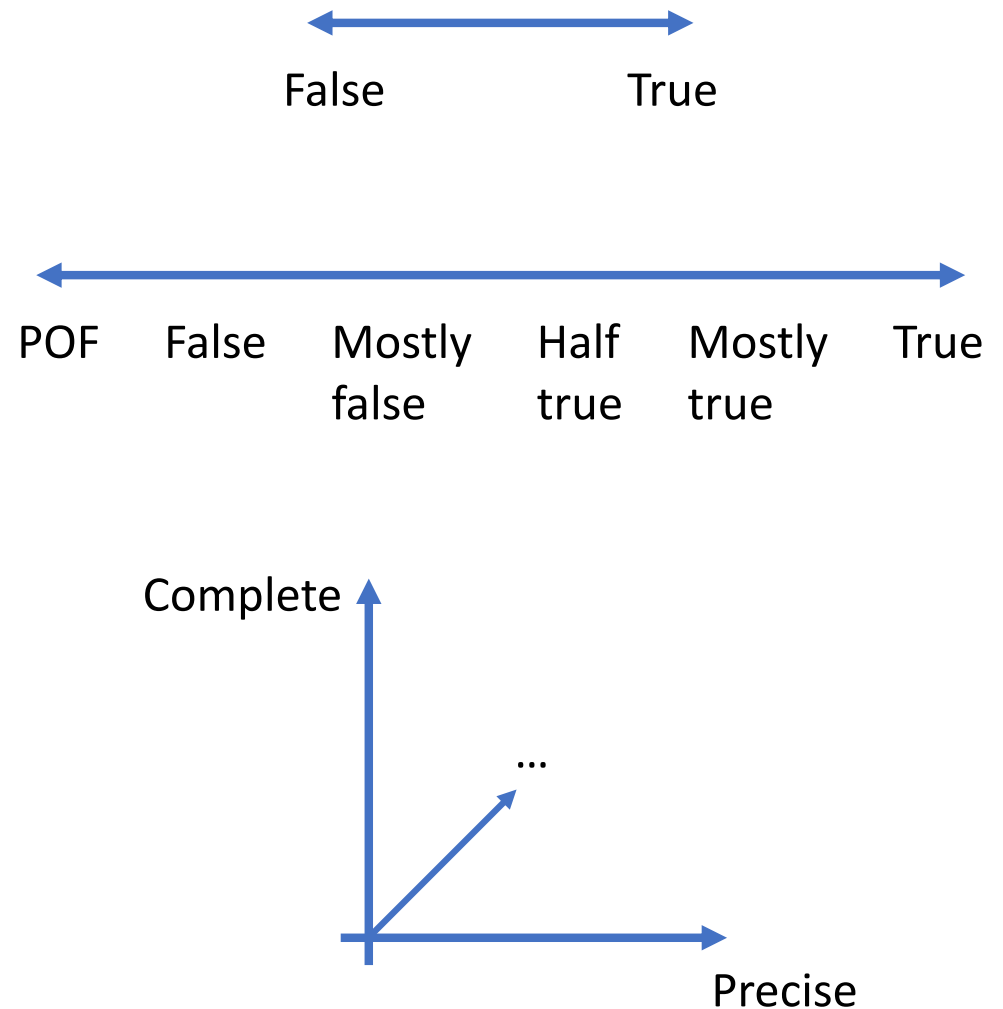
The Poynter Institute

Ma anche

- 100 livelli
- Magnitude estimation (stima della magnitudine)
 - Un numero qualsiasi da zero (escluso) in su
- Preferenze
 - Coppie (A più vero di B)
 - Elenco (ordinare)

And, it's Multidimensional!

- Not only more than two ordinal categories
- Also more than one dimension
- Even more than 2
- We proposed 7
 - → next slides
 - Each on 5 levels [-2,+2]



The 7 dimensions (1/2)

1. **Correctness:** the statement is expressed in an accurate way, as opposed to being incorrect and/or reporting mistaken information
 - Ex. (label: +2 Completely Agree): "It's illegal to treat a minor without parental consent in the U.S. Even as hospitals are limiting visitors, minors will always be allowed to have one"
2. **Neutrality:** the statement is expressed in a neutral / objective way, as opposed to subjective / biased
 - Ex. (label: -2 Completely Disagree): "The Labor Party has repeatedly claimed the Coalition needs to make cuts of \$70 billion to vital services to balance the budget."
3. **Comprehensibility:** the statement is comprehensible / understandable / readable as opposed to difficult to understand
 - Ex. (label: +2 Completely Agree) "Florida ranks first among the nations for access to free prekindergarten."

The 7 dimensions (2/2)

4. **Precision:** the information provided in the statement is precise / specific, as opposed to vague
 - Ex. (label: -2 Completely Disagree): "There were more deaths after the gun bans from guns than there were in the three years before Port Arthur"
5. **Completeness:** the information reported in the statement is complete as opposed to telling only a part of the story
 - Ex. (label: -2 Completely Disagree): "We inherited a broken test for COVID-19."
6. **Speaker's trustworthiness:** the speaker is generally trustworthy / reliable as opposed to untrustworthy / unreliable / malicious
 - Ex. (label: -2 Completely Disagree): "Says video shows the Chinese are destroying the 5G poles as they are aware that it is the thing triggering the corona symptoms.""
7. **Informativeness:** the statement allows us to derive useful information as opposed to simply stating well known facts and/or tautologies.
 - Ex. (label: +2 Completely Agree): "2019 coronavirus can live for up to 3 hours in the air, up to 4 hours on copper, up to 24 hours on cardboard up to 3 days on plastic and stainless steel."

So, Misinformation "definition(s)"

- Terminology (?)
- False information? No! Even true, if misleading
- And, what is false? Against scientific consensus? (no, changing!)
- It's not binary: levels of truthfulness
- It's also multidimensional
- Different kinds of misinformation
- Maybe it's even a strategy
- ... it's complicated

I give up

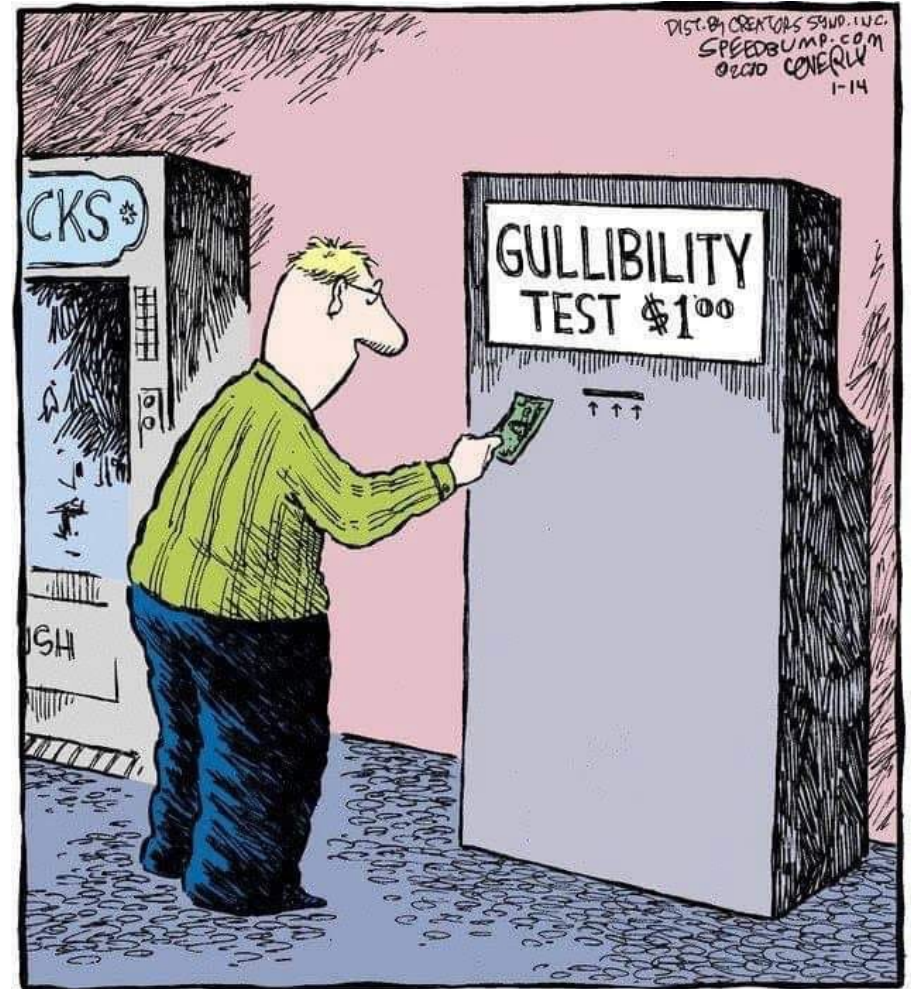
- "The problem is not that we need to have a shared definition of what misinformation is. The problem is that people are assuming that they have a shared definition, and they're just using different definitions."

[Gordon Pennycook, a psychologist studying misinformation at Cornell University]

- Different research(ers) will use different definitions
- Let's acknowledge that and move on

Scaletta

- Intro
 - Terminologia/Definizioni
 - Motivazioni
- Modelli epidemici
- 3 studi sperimentali



Motivazioni

- Perché questo argomento?
- Per Social Computing
 - Dopo i modelli teorici sulla diffusione info delle lezioni scorse, studi sperimentali/pratici
 - Vedete lavori scientifici
 - (che vi "filtro" e semplifico...)
 - Ci torneremo su nella parte sul Crowdsourcing
- Però anche in generale
 - È importante socialmente
 - Alto impatto
 - "Infodemia"
 - È interessante scientificamente
 - Problema/i per cui non ci sono (ancora?) soluzioni

Motivazioni

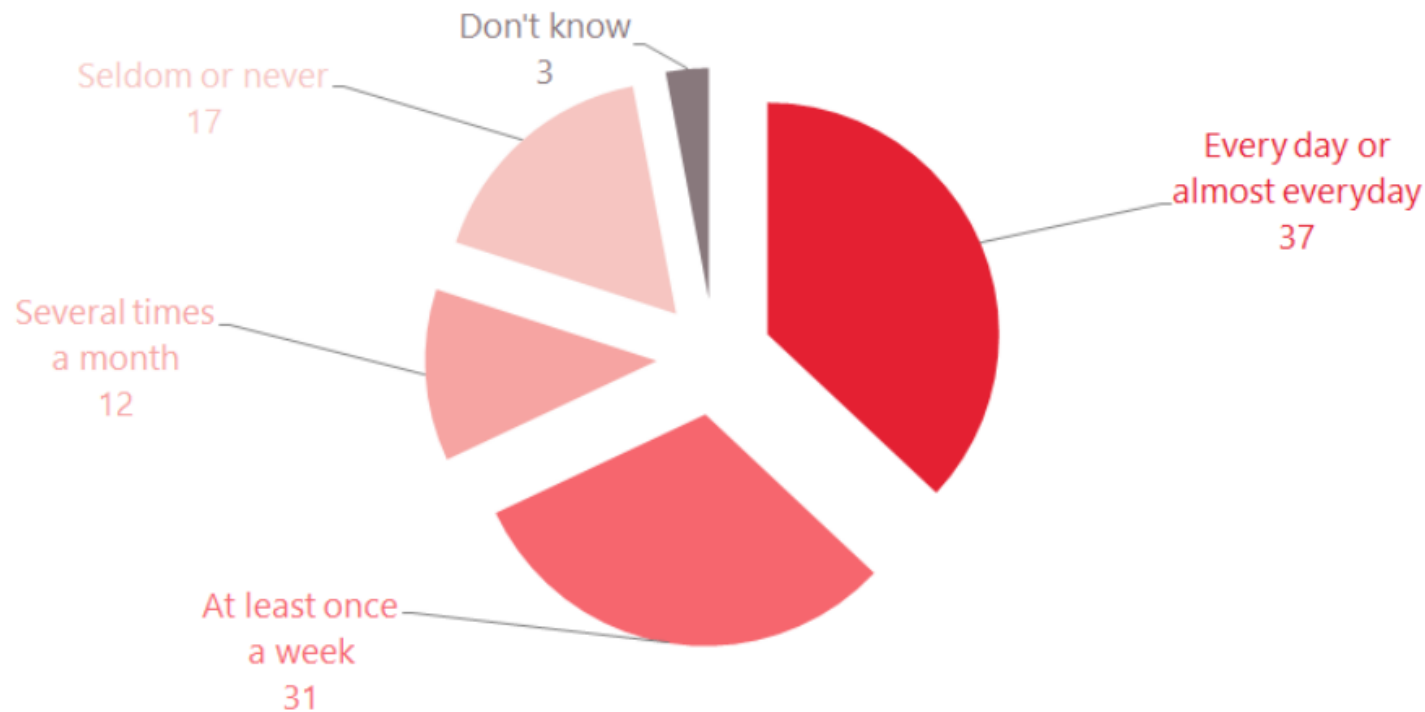
- Perché questo argomento?
- Per Social Computing
 - Dopo i modelli teorici sulla diffusione info delle lezioni scorse, studi sperimentali/pratici
 - Vedete lavori scientifici
 - (che vi "filtro" e semplifico...)
 - Ci torneremo su nella parte sul Crowdsourcing
- Però anche in generale
 - È importante socialmente
 - Alto impatto
 - "Infodemia"
 - È interessante scientificamente
 - Problema/i per cui non ci sono (ancora?) soluzioni

Importanza, percezione cittadini

Is Online Misinformation a Big Problem for Citizens?



Q2 How often do you come across news or information that you believe misrepresent reality or is even false?
(% - EU)



Base: All Respondents (N=26,576)

Source: Eurobarometer 2018



Importanza. "Infodemia"

- «We're concerned about the levels of **rumours** and **misinformation** that are hampering the response. [...] we're not just fighting an epidemic; we're fighting an **infodemic**.
- **Fake news** spreads faster and more easily than this virus, and is just as **dangerous**. That's why we're also working with search and media companies like Facebook, Google, Pinterest, Tencent, Twitter, TikTok, YouTube and others to counter the spread of rumours and misinformation.
- We call on all governments, companies and news organizations to work with us to sound the appropriate level of alarm, without fanning the flames of hysteria.»
 - Dr. Tedros Adhanom Ghebreyesus, the WHO (World Health Organization) Director General, speech at the Munich Security Conference on 15 February 2020
 - <https://www.who.int/dg/speeches/detail/munich-security-conference>

"Infodemia"

• 17/11/2020



«Troppe fake news la pandemia si supera coi risultati scientifici»

Il presidente dell'Iss Brusaferro: l'università è chiamata a investire nella ricerca
E il ministro Manfredi conferma i fondi su diritto allo studio, personale e medicina

Giacomina Pellizzari

UDINE

«La pandemia si supera con le conoscenze scientifiche non con informazioni spesso distorte o comunque non attinenti alla realtà che continuano a circolare attraverso i social network aperti a ognuno di noi». Collegato da remoto, dal salone di palazzo Antonini-Maseri, il presidente dell'Istituto superiore di sanità (Iss), **Silvio Brusaferro**, ha messo in guardia dalle fake news e sollecitato il sistema universitario, compreso l'ateneo friulano, a ripensare le attività di formazione, ricerca e della cosiddetta terza missione in funzione delle sfide future a cui la pandemia ci sta sottoponendo.

La cerimonia di inaugurazione dell'anno accademico, il quarantatreesimo dell'ateneo voluto dalla gente, si è trasformata in un'analisi su come contrastare il coronavirus. «Lo sforzo di traslare la conoscenza nei nostri contesti sociali» rientra nella terza missione che assieme alla formazione e alla ricerca può indicare la strada. Brusaferro, in qualità di componente della comunità accademica friulana e di presidente dell'Iss, ha invitato a riflettere proprio sul «fenomeno dell'infodemia, ovvero sulla circolazione massiva di informazioni non sempre verificate e corrette. Abbiamo tutti testimonianze quotidiane di

in discussione. In questa prospettiva – ha aggiunto – tutto il mondo universitario è chiamato a effettuare sforzi e investimenti straordinari nella ricerca».

Il numero uno dell'Iss è convinto che «riusciremo a superare questa sfida grazie alla ricerca scientifica, che tocca tutte le discipline e che raggiunge i migliori risultati con la condivisione globale dei dati, senza vincoli di confine. La ricerca di cui abbiamo bisogno non è solo quella biomedica – ha concluso –, un po' tutti i settori so-

no chiamati a dare delle risposte». Brusaferro pensa all'informatica, alla sensoristica, alla chimica, alla comunicazione, al diritto, che sono anche i settori su cui sta tarando l'offerta didattica l'università di Udine.

L'altro elemento portante per un Paese – sono sempre le parole di Brusaferro – è la capacità di formare professionisti preparati ad affrontare il Covid e altre eventuali emergenze sanitarie. In questo caso la formazione non può fermarsi a quella universitaria. Brusaferro pensa, infatti, alla formazione continua e alla formazione post-laurea «come elementi da strutturare in modo stabile e sistematico». Da qui l'invito

sionisti», Brusaferro ha concluso augurando alle giovani generazioni che stanno osservando la situazione attuale, di «entusiasmarsi e di decidere di spendere la loro vita nella ricerca per mettere al servizio delle comunità nuove conoscenze e nuovi strumenti». Un tema, questo, affrontato anche dal professor Michele Morgante, ordinario di Genetica, «Dai geni ai genomi, dalla

struttura alla funzione. Una rincorsa fra scienza e tecnologia alla ricerca dell'identità dei viventi».

Anche il ministro dell'Università, **Gaetano Manfredi**, collegato da Roma a pochi minuti dall'inizio del Consiglio dei ministri, si è soffermato sull'importanza della ricerca scientifica. «È intenzione del Governo e del Ministero incidere sugli investimenti per il diritto allo studio, sia utilizzando i fondi ordinari, sia le risorse del Recovery fund, per aumentare il sostegno alla residenzialità, alle borse di studio, e per un'ulteriore riduzione in futuro della tassazione degli studenti universitari». Proprio per invogliare

come circolino e trovino credito le fake news e come i fondamenti della nostra società e i metodi scientifici siano messi

to ad analizzare «il grande esperimento naturale della didattica a distanza che l'università ha saputo mettere in piedi in poco tempo». E nel ringraziare la scuola di medicina dell'ateneo friulano perché ha saputo formare «ottimi profes-



Altri documenti ufficiali

- Tema caldo
 - EU
 - Documenti Horizon Europe, Work programme 2021 – 2022
 - ITA
 - PNR – Programma Nazionale per la Ricerca 2021 – 2027
 - Documenti governativi
- Ricerca scientifica
- Sicurezza



EN

Horizon Europe

Work Programme 2021-2022

6. Civil Security for Society

DISCLAIMER

This draft has not been adopted or endorsed by the European Commission. Any views expressed are the preliminary views of the Commission services and may not in any circumstances be regarded as stating an official position of the Commission. The information transmitted is intended only for the Member State or entity to which it is addressed for discussions and may contain confidential and/or privileged material.

Table of contents

Introduction	8
--------------------	---

Better protect the EU and its citizens against Crime and Terrorism	12
--	----

Call - Fighting Crime and Terrorism 2021	16
--	----

Conditions for the Call	16
-------------------------------	----

FCT01 - Modern information analysis for fighting crime and terrorism	17
--	----

HORIZON-CL3-2021-FCT-01-01: Terrorism and other forms of serious crime countered using travel intelligence	17
--	----

HORIZON-CL3-2021-FCT-01-02: Lawful interception using new and emerging technologies (5G & beyond, quantum computing and encryption)	20
---	----

HORIZON-CL3-2021-FCT-01-03: Disinformation and fake news are combated and trust in the digital world is raised	22
--	----

HORIZON-CL3-2021-FCT-01-04: Improved access to fighting crime and terrorism research data	24
---	----

FCT02 - Improved forensics and lawful evidence collection	27
---	----

HORIZON-CL3-2021-FCT-01-05: Modern biometrics used in forensic science and by police	27
--	----

FCT03 - Enhanced prevention, detection and deterrence of societal issues related to various forms of crime	29
--	----

HORIZON-CL3-2021-FCT-01-06: Domestic and sexual violence are prevented and combated	29
---	----

FCT04 - Increased security of citizens against terrorism, including in public spaces	32
--	----

HORIZON-CL3-2021-FCT-01-07: Improved preparedness on attacks to public spaces.....	33
--	----

FCT05 - Organised crime prevented and combated.....	35
---	----

HORIZON-CL3-2021-FCT-01-08: Fight against trafficking in cultural goods	35
---	----

HORIZON-CL3-2021-FCT-01-09: Fight against organised environmental crime	38
---	----

HORIZON-CL3-2021-FCT-01-10: Fight against firearms trafficking.....	41
---	----

FCT06 – Citizens are protected against cybercrime	44
---	----

HORIZON-CL3-2021-FCT-01-11: Prevention of child sexual exploitation.....	44
--	----

HORIZON-CL3-2021-FCT-01-12: Online identity theft is countered	46
--	----

Call - Fighting Crime and Terrorism 2022	48
--	----

Conditions for the Call	48
-------------------------------	----

FCT02 - Improved forensics and lawful evidence collection	49
---	----

HORIZON-CL3-2022-FCT-01-01: Improved crime scene investigations related to transfer, persistence and background abundance	50
---	----

HORIZON-CL3-2022-FCT-01-02: Better understanding the influence of organisational cultures and human interactions in the forensic context as well as a common lexicon	52
--	----

FCT03 Enhanced prevention, detection and deterrence of societal issues related to various forms of crime	54
--	----



PNR 2021-2027
Programma nazionale
per la ricerca

all'economia e mettono in pericolo l'incolumità dei cittadini quando colpiscono infrastrutture critiche e reti di distribuzione dei servizi essenziali come la sanità, l'energia, gli acquedotti, i trasporti. In Italia, interi settori di eccellenza, come la meccanica, la cantieristica, il Made in Italy, il turismo, l'agroalimentare, il farmaceutico e i trasporti, potrebbero subire pesanti ridimensionamenti di fatturato a causa di attacchi perpetrati da stati sovrani o da concorrenti sleali. Oggetto di attacco può essere anche il sistema politico e con esso la democrazia e i processi decisionali che ne sono alla base. La storia recente dimostra come campagne di *fake news* possano orientare l'opinione politica degli elettori, condizionando l'esito di elezioni e processi di governance. È necessario mettere a punto sistemi di difesa, coordinati a livello nazionale, per contrastare le minacce e, ove possibile, prevenirle. Per questo la ricerca di base e applicata in *cybersecurity* deve essere considerata una priorità per il Paese. Il radicale cambiamento di architettura delle infrastrutture di comunicazione e di erogazione dei servizi, conseguente alla diffusione su larga scala delle tecnologie IoT, del 5G, dell'intelligenza artificiale e delle tecnologie quantistiche, introdurrà nuove criticità dovute al progressivo rilassamento del concetto di "perimetro di difesa" e al conseguente crollo di buona parte delle strategie di difesa e protezione che si basano su tale concetto.

La *cybersecurity* non è solo la definizione di strategie di difesa atte a contrastare gli attacchi, mitigare il rischio e

Articolazione 4. Sicurezza dei servizi al cittadino e alle imprese

L'uso di sistemi informatici sofisticati è ormai diventato ineludibile per l'interazione del cittadino con le istituzioni pubbliche (*e-government*), con soggetti privati, con le cose (IoT), con l'ambiente (*smart cities*). Tuttavia, i sistemi attuali non garantiscono un buon controllo dei processi e neppure la possibilità di una verifica trasparente dei dati coinvolti. Pertanto, è necessario sviluppare strumenti e metodologie che rendano i processi trasparenti, verificabili e decentralizzati e fondati su nuovi modelli di *trust*. Sono numerosi i contesti nei quali è necessario tracciare processi complessi che coinvolgono molteplici entità. Per difendere il Made in Italy in importanti sistemi produttivi, quali l'agroalimentare, l'abbigliamento, il farmaceutico, è importante definire strumenti e sistemi di tracciabilità e comunicazione trasparente. Altrettanto importante è la protezione del diritto d'autore, che possa essere realizzata in maniera decentralizzata attraverso l'interazione diretta tra creatore e fruitore della stessa. La tecnologia delle blockchain risulta molto promettente a questo scopo e la disponibilità di una infrastruttura di blockchain nazionale rappresenterebbe una garanzia e uno stimolo per tutti i soggetti coinvolti. Anche la qualità dei servizi sanitari potrebbe beneficiare di piattaforme che facilitino convergenza e accessibilità a dati sanitari da remoto in tempo reale e sicuro, ma è altrettanto importante offrire al cittadino la possibilità di verificare l'uso dei dati raccolti. Un'altra area rilevante è quella relativa alla definizione di strumenti e di tecniche per l'analisi, la tracciabilità e la verificabilità di notizie provenienti da media/reti sociali per contrastare tecniche di misinformazione attraverso le *fake news* e i *deep fakes*.



Ministero dell'Università e della Ricerca

Intelligence, Cybersecurity, Fake news

- "Minaccia ibrida"
 - (pubblicato in febbraio 2021)



MINACCIA IBRIDA

in breve

- Con la pandemia, impennata di campagne disinformative e fake news
- Dilatati margini di intervento per attori ostili propensi all'uso combinato di più strumenti a fini manipolatori e d'influenza
- Nuovi indirizzi operativi della UE

Mirata e coordinata azione intelligence è stata riservata alla cd. minaccia ibrida – per definizione veicolata su diversi domini (quello diplomatico, militare, economico/finanziario, intelligence, etc.) – che, in concomitanza con il dispiegarsi dell'emergenza sanitaria, è stata caratterizzata da costanti tentativi di intossicazione del dibattito pubblico attraverso attività di disinformazione e/o di influenza, nel contesto di più ampie campagne ibride. Al riguardo, è stata registrata una elevatissima produzione di fake news e narrazioni allarmistiche, sfociate in un surplus informativo (cd. infodemia) di difficile discernimento per la collettività. Fattore di rischio intrinseco al fenomeno della disinformazione online ha continuato a risiedere nelle logiche e negli algoritmi alla base dello stesso funzionamento dei social media, tendenti a creare un ambiente autoreferenziale ed autoalimentante, fondato sulla condivisione dei contenuti e delle relazioni di interesse che, polarizzando l'informazione disponibile, ne alimenta quindi la percezione parziale e faziosa.

15. Diffusione disinformazioni - S. Mizzaro



PRESIDENZA DEL CONSIGLIO DEI MINISTRI



SISTEMA DI INFORMAZIONE
PER LA SICUREZZA DELLA REPUBBLICA

RELAZIONE SULLA POLITICA DELL'INFORMAZIONE PER LA SICUREZZA 2020



PALAZZO DANTE

Anche nel 2022

- (pubblicato in febbraio 2022)

Nell'ambito della minaccia endogena relativa a eversione ed estremismi, l'intelligence si è impegnata a cogliere segnali e trend della conflittualità sociale in relazione sia al consueto attivismo dell'oltranzismo politico di diversa matrice, sia a quelle peculiari e composite espressioni del dissenso che, germinate essenzialmente sul web, hanno infiammato, all'insegna dell'opposizione alle misure governative di contenimento dei contagi, diverse piazze italiane. A questo riguardo, si è assistito, al pari di altri Paesi europei, all'evolversi di una spirale mobilitativa che ha trovato linfa vitale in un ingente flusso virtuale di **fake news** e teorie cospirative sull'emergenza pandemica.

In tale contesto, l'attività di ricerca informativa ha consentito di rilevare nel periodo di riferimento campagne di **disinformazione** condotte in parallelo a offensive digitali in concomitanza con l'emergenza sanitaria, nel tentativo di incidere sul tessuto sociale, facendo leva sull'emotività dell'opinione pubblica, al fine di ottenere un vantaggio strategico.



Barack Obama

- Discorso a Stanford
 - 21/4/2022
- Guardatelo!

- <https://youtu.be/ExEApwbhfqQ>



Barack Obama

Apr 22 · 28 min read · Listen



Disinformation Is a Threat to Our Democracy

Tech platforms need to recognize that their decisions have an impact on every aspect of society.



Un articolo recente...

- Budak, C., Nyhan, B., Rothschild, D.M., Thorson, E. & Watts, D.J. Misunderstanding the harms of online misinformation. *Nature* **630**, 45–53 (June 2024). <https://doi.org/10.1038/s41586-024-07417-w>
- "Perspective" → opinione
 - In realtà non è così diffuso/grave...

Perspective

Misunderstanding the harms of online misinformation

<https://doi.org/10.1038/s41586-024-07417-w>

Received: 13 October 2021

Accepted: 11 April 2024

Published online: 5 June 2024

 Check for updates

Ceren Budak¹, Brendan Nyhan², David M. Rothschild^{3,4,5}, Emily Thorson⁴ & Duncan J. Watts⁵

The controversy over online misinformation and social media has opened a gap between public discourse and scientific research. Public intellectuals and journalists frequently make sweeping claims about the effects of exposure to false content online that are inconsistent with much of the current empirical evidence. Here we identify three common misperceptions: that average exposure to problematic content is high, that algorithms are largely responsible for this exposure and that social media is a primary cause of broader social problems such as polarization. In our review of behavioural science research on online misinformation, we document a pattern of low exposure to false and inflammatory content that is concentrated among a narrow fringe with strong motivations to seek out such information. In response, we recommend holding platforms accountable for facilitating exposure to false and extreme content in the tails of the distribution, where consumption is highest and the risk of real-world harm is greatest. We also call for increased platform transparency, including collaborations with outside researchers, to better evaluate the effects of online misinformation and the most effective responses to it. Taking these steps is especially important outside the USA and Western Europe, where research and data are scant and harms may be more severe.

Social media platforms have become ubiquitous in modern life, prompting great interest in their effects on society. Debates over social media's impact often include claims that social media platforms and algorithms facilitate high levels of exposure to misinformation and other harmful content¹, which in turn cause social problems ranging from polarization to political violence^{2–3}. Broad causal claims like these appear frequently in influential public discourse about social media, including in articles written by prominent social scientists⁴. For example, a widely discussed article in *The Atlantic* argues that “social media amplifies political polarization; foments populism, especially right-wing populism; and is associated with the spread of misinformation”⁴. Similarly, *The New York Times* op-eds have argued that “YouTube may be one of the most powerful radicalizing instruments of the 21st century” and that “The dominant digital platform companies, including Facebook and Google ... have created a haven for dangerous misinformation and hate speech that has undermined trust in democratic institutions”^{4,5}. These concerns are echoed by both politicians⁶ and members of the US public^{7–9}, who say that social media has a negative impact on society and cite misinformation as a primary reason¹⁰.

Assertions like these suggest that exposure to false content on social media has been shown to cause massive social harms. However, we do not believe that compelling empirical evidence exists supporting these claims. The question of whether and when exposure to this type of content on social media causes harm is complex, a matter of active debate among researchers and extremely difficult to conclusively answer. Although disjunctures between public discourse and scholarly work

are a problem in many fields^{11–14}, we argue that unsupported or disputed claims about widespread exposure to (and effects of) misinformation on social media are particularly concerning because they can shape the actions of platforms, legislators and regulators, diverting attention from other ways in which misinformation can cause harm.

Before proceeding, we note that this Perspective focuses strictly on the potential effects of exposure to false and inflammatory content on social media. We acknowledge that reasons for concern exist about other potential harms from social media such as negative effects on mental health and well-being^{15,16} and social trust^{17,18} (see Table 1 for a summary or ref. 19 for a review). Although these other potential harms are also important, concerns about misinformation have been a focus of much of the public discourse around social media. In addition, narrowing our focus allows for a more thorough review of the relevant literature.

The public debate that we engage with concerns the effects on mass opinion of exposure to misinformation and extremism (for example, white supremacist, alt-right) content on social media. That debate often centres on three claims that are largely unsupported by academic research: (1) average exposure to misinformation is high and growing; as a result, a substantial fraction of the population is exposed to it frequently (for example, refs. 20–24); (2) exposure to this content is primarily driven by the platforms' algorithms rather than by individual users deliberately seeking out such content (for example, refs. 1, 3, 25–28); and (3) correlations between exposure to false and extremist content on social media platforms and undesirable

¹University of Michigan School of Information, Ann Arbor, MI, USA. ²Department of Government, Dartmouth College, Hanover, NH, USA. ³Microsoft Research, New York, NY, USA. ⁴Maxwell School of Citizenship and Public Affairs, Syracuse University, Syracuse, NY, USA. ⁵Department of Computer and Information Science, Annenberg School of Communication, and Operations, Information, and Decisions Department, University of Pennsylvania, Philadelphia, PA, USA. [✉]e-mail: david@researchdmr.com

Commenti

- Sulla base di una rassegna dei risultati in letteratura
- «three common **misperceptions**:
 - average exposure to problematic content is high
 - algorithms are largely responsible for this exposure
 - social media is a primary cause of broader social problems such as polarization.»

Commenti

- Sulla base di una rassegna dei risultati in letteratura
- «three common **misperceptions**:
 - average exposure to problematic content is high
 - algorithms are largely responsible for this exposure
 - social media is a primary cause of broader social problems such as polarization.»
- → C'è discussione su "quanto grave" sia il fenomeno della disinformazione

... e un altro articolo recente...

- Ecker UKH, Tay LQ, Roozenbeek J, van der Linden S, Cook J, Oreskes N, Lewandowsky S. Why misinformation must not be ignored. *Am Psychol.* 2025 Sep;80(6):867-878. doi: 10.1037/amp0001448. Epub 2024 Dec 12. PMID: 39666498.
- No in realtà è diffuso/grave



© 2024 The Author(s)
ISSN: 0003-066X

American Psychologist

2025, Vol. 80, No. 6, 867–878
<https://doi.org/10.1037/amp0001448>

Why Misinformation Must Not Be Ignored

Ullrich K. H. Ecker^{1, 2}, Li Qian Tay³, Jon Roozenbeek^{4, 5}, Sander van der Linden⁵, John Cook⁶, Naomi Oreskes⁷, and Stephan Lewandowsky^{8, 9}

¹ School of Psychological Science, University of Western Australia

² Public Policy Institute, University of Western Australia

³ School of Medicine and Psychology, Australian National University

⁴ Department of War Studies, King's College London

⁵ Department of Psychology, University of Cambridge

⁶ Melbourne Centre for Behavior Change, University of Melbourne

⁷ Department of the History of Science, Harvard University

⁸ School of Psychological Science, University of Bristol

⁹ Department of Psychology, University of Potsdam

Recent academic debate has seen the emergence of the claim that misinformation is not a significant societal problem. We argue that the arguments used to support this minimizing position are flawed, particularly if interpreted (e.g., by policymakers or the public) as suggesting that misinformation can be safely ignored. Here, we rebut the two main claims, namely that misinformation is not of substantive concern (a) due to its low incidence and (b) because it has no causal influence on notable political or behavioral outcomes. Through a critical review of the current literature, we demonstrate that (a) the prevalence of misinformation is nonnegligible if reasonably inclusive definitions are applied and that (b) misinformation has causal impacts on important beliefs and behaviors. Both scholars and policymakers should therefore continue to take misinformation seriously.

Public Significance Statement

This article refutes claims that misinformation is an insignificant issue. Through a critical literature review, we demonstrate that misinformation represents a nontrivial part of the information environment and can causally and adversely influence people's beliefs, decisions, and behaviors. We clarify through our discussion why misinformation continues to be a significant problem that should not be ignored by communicators and policymakers.

Keywords: misinformation, disinformation, information environment, informational influence, public discourse

This article was published Online First December 12, 2024.

Natalia Zarzeczna served as action editor.

Ullrich K. H. Ecker <https://orcid.org/0000-0003-4743-313X>

Li Qian Tay <https://orcid.org/0000-0001-7990-0455>

Jon Roozenbeek <https://orcid.org/0000-0002-8150-9305>

Sander van der Linden <https://orcid.org/0000-0002-0269-1744>

John Cook <https://orcid.org/0000-0001-8488-6766>

Naomi Oreskes <https://orcid.org/0000-0001-6122-9843>

Stephan Lewandowsky <https://orcid.org/0000-0003-1655-2013>

The authors declare no conflicts of interest. Ullrich K. H. Ecker was supported by Australian Research Council Grant FT190100708. Jon Roozenbeek acknowledges funding from the British Academy (Grant PF21 \210010). Sander van der Linden acknowledges funding from the U.K. Government (Grant SCH-00001-3391), Google, and APA/CDC. Stephan Lewandowsky acknowledges support from the European Research Council (ERC Advanced Grant 101020961 PRODEMINFO), the Humboldt

Foundation, the Volkswagen Foundation (grant "Reclaiming Individual Autonomy and Democratic Discourse Online: How to Rebalance Human and Algorithmic Decision Making"), and the European Commission (Horizon 2020 Grant 101094752 SoMe4Dem), Google Jigsaw, and U.K. Research and Innovation (EU Horizon replacement funding Grant 10049415). Naomi Oreskes receives royalties from books and lectures dealing with misinformation and consults to private law firms. The funding sources had no other role other than financial support.

Open Access funding provided by the European Commission and U.K. Research and Innovation: This work is licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0; <http://creativecommons.org/licenses/by/4.0>). This license permits copying and redistributing the work in any medium or format, as well as adapting the material for any purpose, even commercially.

Ullrich K. H. Ecker played a lead role in conceptualization, project administration, and writing—original draft. Li Qian Tay played a supporting

(Opinione personale

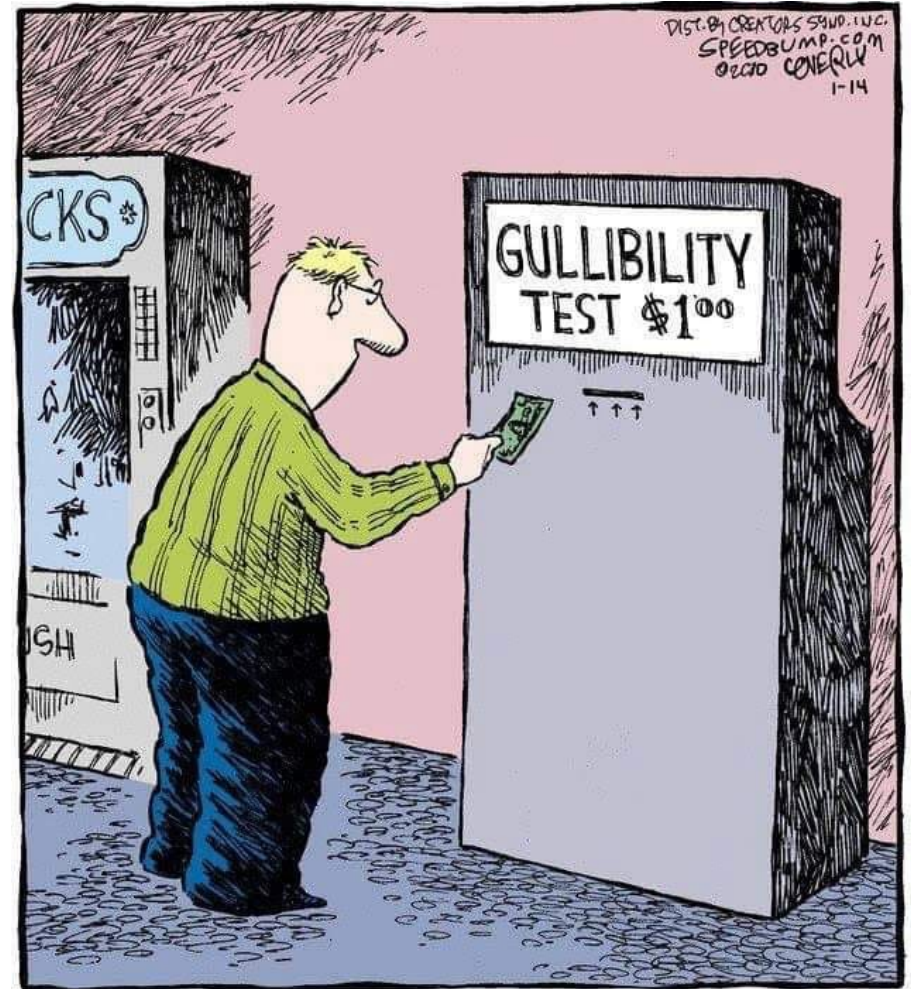
- Io sono preoccupato/spaventato
- Le "AI" possono
 - Creare inavvertitamente disinformazione ("allucinazioni")
 - Essere usate da malintenzionati per creare disinformazione su larga scala
 - Come sarebbe il Web se il 99% delle informazioni fossero false?
 - Creare disinformazione anche multimediale
 - Immagini
 - Video (deepfake)
-)

Insomma

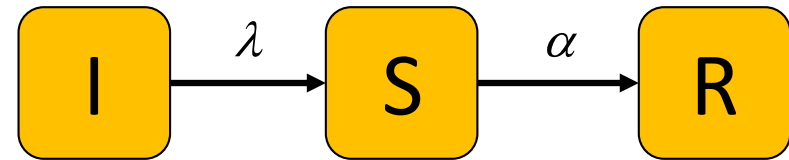
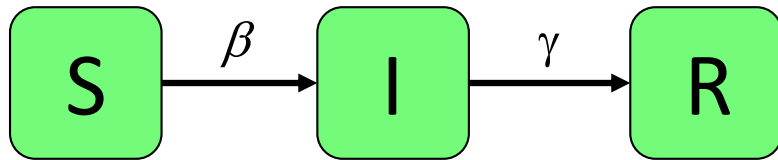
- Cittadini, società ✓
- Stampa ✓
- Politica, ... ✓
- Le "fake news" sono un problema, da combattere
 - Anche se qualcuno sostiene che sia meno grave di quanto percepito
- Vediamo cosa dice La Scienza
 - Modelli epidemici
 - Tre studi sperimentali
 - (c'è moooooooooooooooooolto altro!!)

Scaletta

- Intro
 - Terminologia/Definizioni
 - Motivazioni
- Modelli epidemici
- 3 studi sperimentali

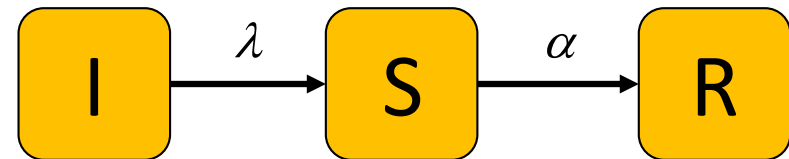
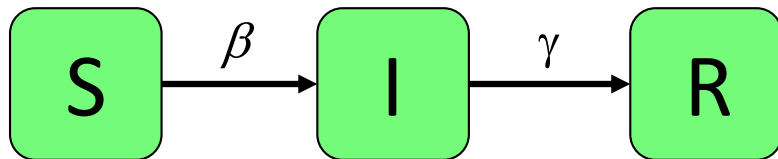


Modelli epidemici per disinformazione?

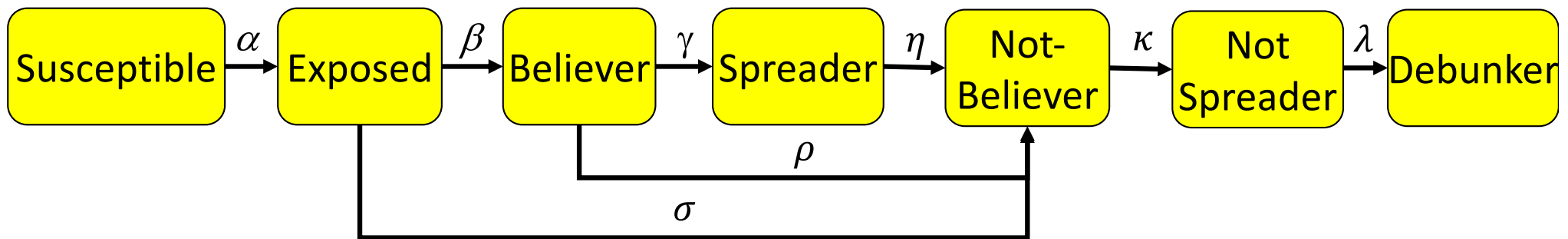


- Abbiamo visto SIR / ISR
- Non chiaro quale dei due, non chiaro significato dei compartimenti
 - Chi non ha ricevuto fake news: **Susceptible?** Ignorant?
 - Chi l'ha ricevuta e la distribuisce: Infected? **Spreader?**
 - Chi non la distribuisce più: **Recovered?** Stifler?
 - E poi:
 - Chi non la distribuisce, passivamente (Recovered?) vs.
 - Chi fa Debunking, la ostacola attivamente (Stifler?)

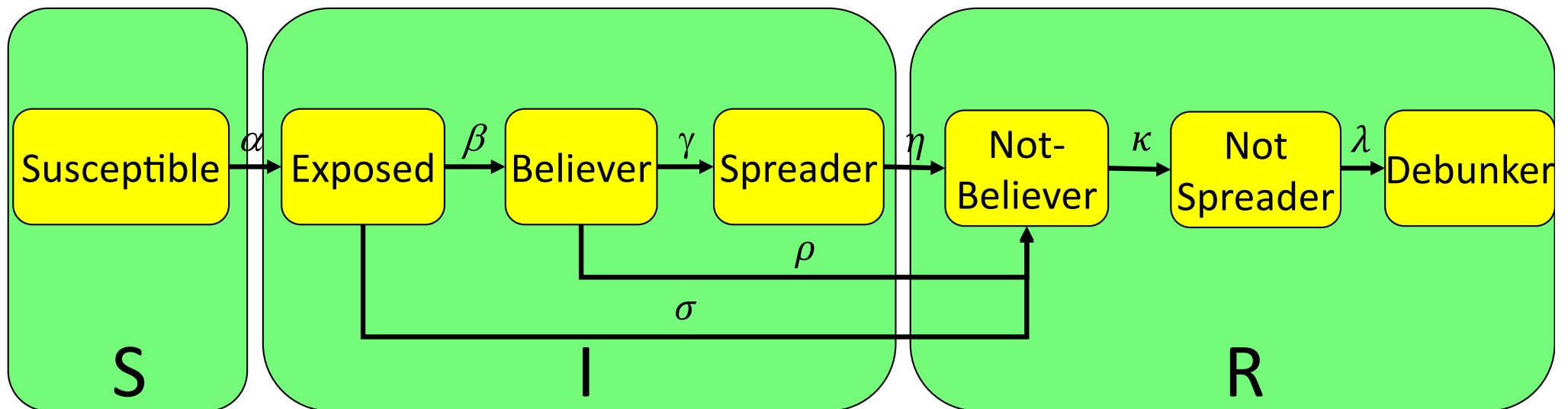
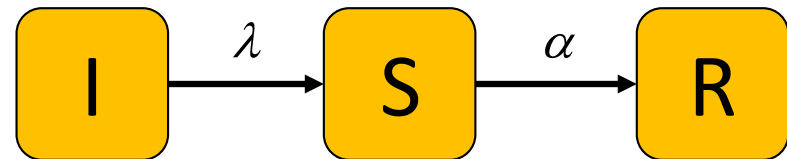
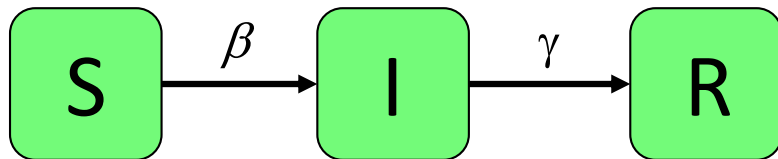
Modelli epidemici per disinformazione?



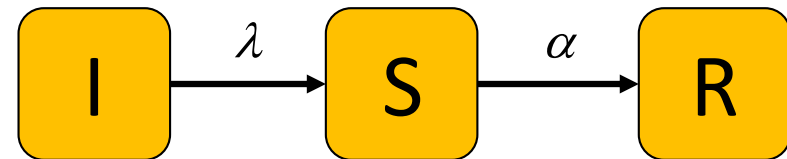
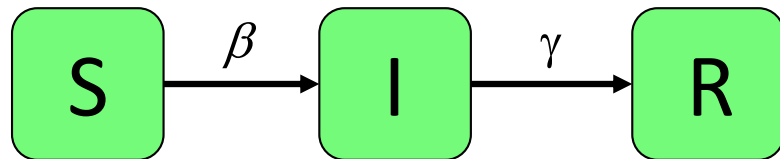
- Forse per la disinformazione ha senso aggiungere qualche compartimento
- Una possibilità:



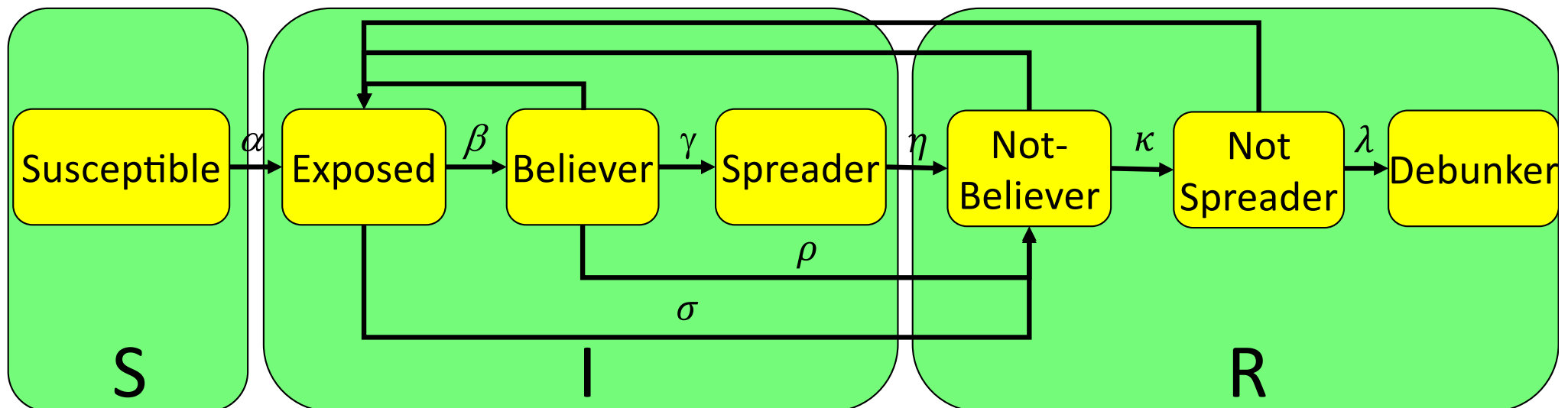
Confronto con SIR



E ci sono altre transizioni!

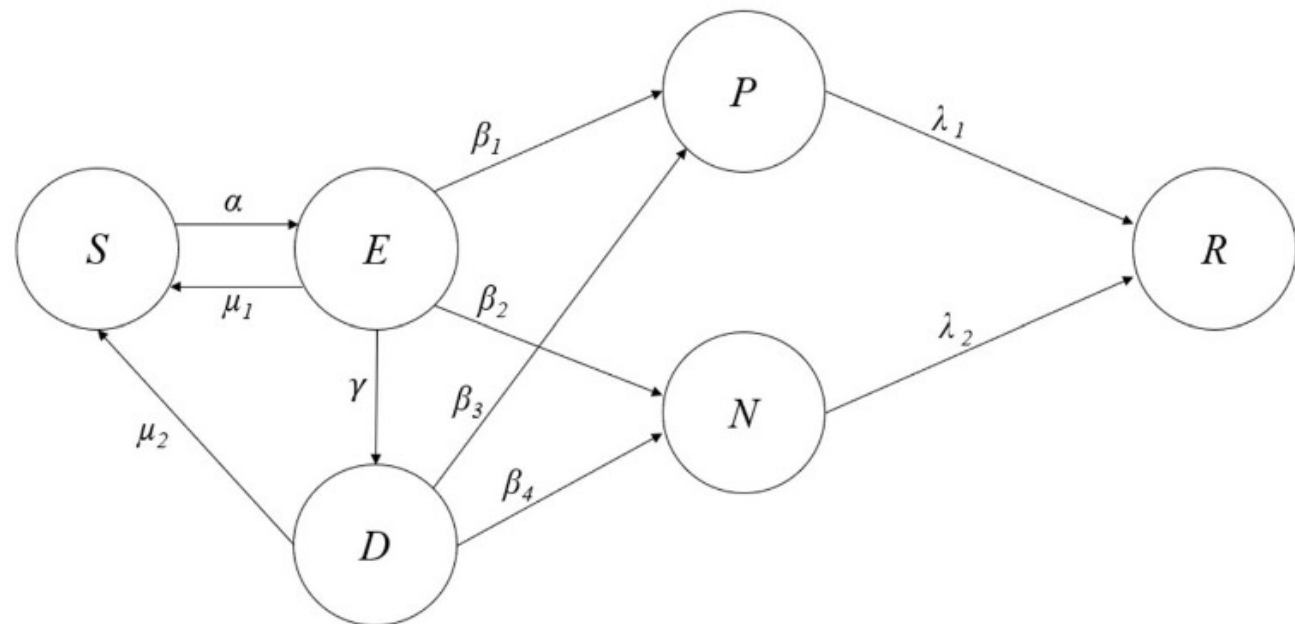


- (e altre ancora, probabilmente...)



Un modello analogo

- SEDPNR (Susceptible – Exposed – Doubtful – Positively Infected – Negatively Infected – Restrained)
- Govindankutty, S., Gopalan, S.P. Epidemic modeling for misinformation spread in digital networks through a social intelligence approach. *Sci Rep* **14**, 19100 (2024) <https://doi.org/10.1038/s41598-024-69657-0>



Insomma

- "M/Disinformazione come virus"
- Idea di base (modello SIR/ISR-like per misinformazione)
- Adattabile
- Con un po' di sforzo...
- (ci sono anche proposte più simili al modello di Morris, basate su teoria dei giochi)

"Vaccini" per disinformazione

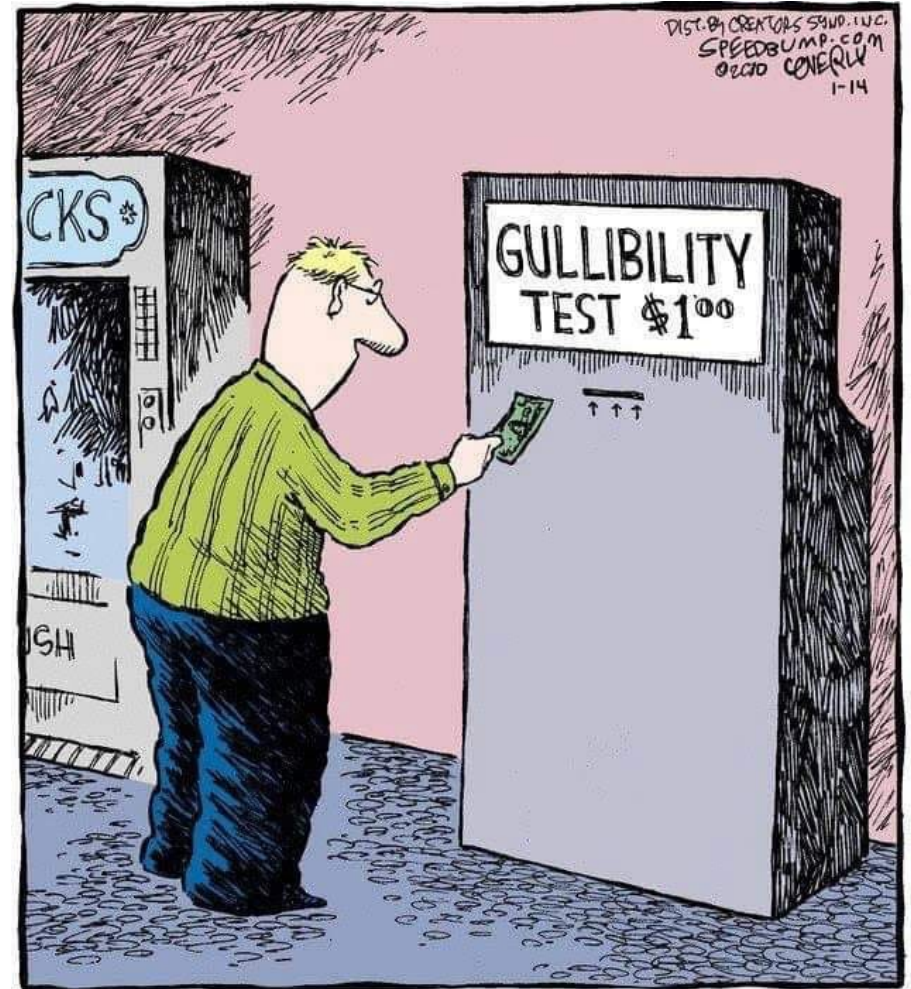
- Altra similitudine Epidemia – Infodemia
- Mostrare info "falsa ma non pericolosa", "leggermente falsa", ecc. per aumentare la resistenza a fake news future
- 2 tipi di "Inoculation":
 - «(1) issue-based, inoculating people against specific examples of misinformation, or
 - (2) technique-based, focusing on manipulation strategies that are often used to mislead people»

Articoli su "Inoculation"

- Kai Kupferschmidt, You won't believe this, *Science*, Vol. 386 Issue 6721, 1 November 2024
 - <https://www.science.org/content/article/can-people-be-inoculated-against-misinformation>
- Sander van der Linden, Jon Roozenbeek, "Inoculation" to Resist Misinformation, *JAMA*, Volume 331, Issue 22
 - https://scholar.google.com/citations?view_op=view_citation&hl=it&user=gUnA_QQAAAAJ&sortby=pubdate&citation_for_view=gUnA_QQAAAAJ:OBsAB-F7qqsC

Scaletta

- Intro
 - Terminologia/Definizioni
 - Motivazioni
- Modelli epidemici
- 3 studi sperimentali



Tre studi

- Studi sperimentali sulla diffusione delle disinformazioni (aka "fake news")



I tre studi

- [Science] Soroush Vosoughi, Deb Roy, Sinan Aral (2018). The spread of true and false news online. *Science*, 359(6380), 1146-1151. DOI: 10.1126/science.aap9559
<https://science.sciencemag.org/content/359/6380/1146>
- [PNAS] Michela Del Vicario, Alessandro Bessi, Fabiana Zollo, Fabio Petroni, Antonio Scala, Guido Caldarelli, H. Eugene Stanley, and Walter Quattrociocchi (2016). The spreading of misinformation online. *PNAS* 2016 113(3)
<https://doi.org/10.1073/pnas.1517441113>
- [PLoS ONE] Fabiana Zollo, Alessandro Bessi, Michela Del Vicario, Antonio Scala, Guido Caldarelli, Louis Shekhtman, Shlomo Havlin, Walter Quattrociocchi (2017) Debunking in a world of tribes. *PLoS ONE* 12(7): e0181821.
<https://doi.org/10.1371/journal.pone.0181821>
 - (con eventuali "Supplementary material")

Primo studio [Science]

- Twitter
- Diffusione
- ~ 126 000 storie ("rumors") dal 2006 al 2017
 - Insieme di tweet/retweet
- 3 milioni di utenti
- 4.5 milioni di tweet
- Classificate in vere e false
- Usando servizi terze parti:
 - snopes.com
 - politifact.com
 - factcheck.org
 - truthorfiction.com
 - hoax-slayer.com
 - urbanlegends.about.com

15. Diffusione disinformazioni - S. Mizzaro

RESEARCH

SOCIAL SCIENCE

The spread of true and false news online

Soroush Vosoughi,¹ Deb Roy,¹ Sinan Aral^{2*}

We investigated the differential diffusion of all of the verified true and false news stories distributed on Twitter from 2006 to 2017. The data comprise ~126,000 stories tweeted by ~3 million people more than 4.5 million times. We classified news as true or false using information from six independent fact-checking organizations that exhibited 95 to 98% agreement on the classifications. Falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information, and the effects were more pronounced for false political news than for false news about terrorism, natural disasters, science, urban legends, or financial information. We found that false news was more novel than true news, which suggests that people were more likely to share novel information. Whereas false stories inspired fear, disgust, and surprise in replies, true stories inspired anticipation, sadness, joy, and trust. Contrary to conventional wisdom, robots accelerated the spread of true and false news at the same rate, implying that false news spreads more than the truth because humans, not robots, are more likely to spread it.

Foundational theories of decision-making (1–3), cooperation (4), communication (5), and markets (6) all view some conceptualization of truth or accuracy as central to the functioning of nearly every human endeavor. Yet, both true and false information spreads rapidly through online media. Defining what is true and false has become a common political strategy, replacing debates based on a mutually agreed set of facts. Our economies are not immune to the spread of falsity either. False rumors have affected stock prices and the motivation for large-scale investments, for example, wiping out \$130 billion in stock value after a false tweet claimed that Barack Obama was injured in an explosion (7). Indeed, our responses to everything from natural disasters (8, 9) to terrorist attacks (10) have been disrupted by the spread of false news online.

New social technologies, which facilitate rapid information sharing and large-scale information cascades, can enable the spread of misinformation (i.e., information that is inaccurate or misleading). But although more and more of our access to information and news is guided by these new technologies (11), we know little about their contribution to the spread of falsity online. Though considerable attention has been paid to anecdotal analyses of the spread of false news by the media (12), there are few large-scale empirical investigations of the diffusion of misinformation or its social origins. Studies of the spread of misinformation are currently limited to analyses of small, ad hoc samples that ignore two of the most important scientific questions: How do truth and falsity diffuse differently, and what factors of human judgment explain these differences?

Current work analyzes the spread of single rumors, like the discovery of the Higgs boson (13) or the Haitian earthquake of 2010 (14), and multiple rumors from a single disaster event, like the Boston Marathon bombing of 2013 (10), or it develops theoretical models of rumor diffusion (15), methods for rumor detection (16), credibility evaluation (17, 18), or interventions to curtail the spread of rumors (19). But almost no studies comprehensively evaluate differences in the spread of truth and falsity across topics or examine why false news may spread differently than the truth. For example, although Del Vicario *et al.* (20) and Bessi *et al.* (21) studied the spread of scientific and conspiracy-theory stories, they did not evaluate their veracity. Scientific and conspiracy-theory stories can both be either true or false, and they differ on stylistic dimensions that are important to their spread but orthogonal to their veracity. To understand the spread of false news, it is necessary to examine diffusion after differentiating true and false scientific stories and true and false conspiracy-theory stories and controlling for the topical and stylistic differences between the categories themselves. The only study to date that segments rumors by veracity is that of Friggeri *et al.* (19), who analyzed ~4000 rumors spreading on Facebook and focused more on how fact checking affects rumor propagation than on how falsity diffuses differently than the truth (22).

In our current political climate and in the academic literature, a fluid terminology has arisen around “fake news,” foreign interventions in U.S. politics through social media, and our understanding of what constitutes news, fake news, false news, rumors, rumor cascades, and other related terms. Although, at one time, it may have been appropriate to think of fake news as referring to the veracity of a news story, we now believe that this phrase has been irredeemably polarized in our current political and media climate. As politicians have implemented a political strategy of labeling news sources that do not

support their positions as unreliable or fake news, whereas sources that support their positions are labeled reliable or not fake, the term has lost all connection to the actual veracity of the information presented, rendering it meaningless for use in academic classification. We have therefore explicitly avoided the term fake news throughout this paper and instead use the more objectively verifiable terms “true” or “false” news. Although the terms fake news and misinformation also imply a willful distortion of the truth, we do not make any claims about the intent of the purveyors of the information in our analyses. We instead focus our attention on veracity and stories that have been verified as true or false.

We also purposefully adopt a broad definition of the term news. Rather than defining what constitutes news on the basis of the institutional source of the assertions in a story, we refer to any asserted claim made on Twitter as news (we defend this decision in the supplementary materials section on “reliable sources,” section S1.2). We define news as any story or claim with an assertion in it and a rumor as the social phenomena of a news story or claim spreading or diffusing through the Twitter network. That is, rumors are inherently social and involve the sharing of claims between people. News, on the other hand, is an assertion with claims, whether it is shared or not.

A rumor cascade begins on Twitter when a user makes an assertion about a topic in a tweet, which could include written text, photos, or links to articles online. Others then propagate the rumor by retweeting it. A rumor’s diffusion process can be characterized as having one or more cascades, which we define as instances of a rumor-spreading pattern that exhibit an unbroken retweet chain with a common, singular origin. For example, an individual could start a rumor cascade by tweeting a story or claim with an assertion in it, and another individual could independently start a second cascade of the same rumor (pertaining to the same story or claim) that is completely independent of the first cascade, except that it pertains to the same story or claim. If they remain independent, they represent two cascades of the same rumor. Cascades can be as small as size one (meaning no one retweeted the original tweet). The number of cascades that make up a rumor is equal to the number of times the story or claim was independently tweeted by a user (not retweeted). So, if a rumor “A” is tweeted by 10 people separately, but not retweeted, it would have 10 cascades, each of size one. Conversely, if a second rumor “B” is independently tweeted by two people and each of those two tweets is retweeted 100 times, the rumor would consist of two cascades, each of size 100.

Here we investigate the differential diffusion of true, false, and mixed (partially true, partially false) news stories using a comprehensive data set of all of the fact-checked rumor cascades that spread on Twitter from its inception in 2006 to 2017. The data include ~126,000 rumor cascades spread by ~3 million people more than 4.5 million times. We sampled all rumor cascades investigated by six independent fact-checking organizations

¹Massachusetts Institute of Technology (MIT), the Media Lab, E14-526, 75 Amherst Street, Cambridge, MA 02142, USA. ²MIT, E62-364, 100 Main Street, Cambridge, MA 02142, USA.
*Corresponding author. Email: sinan@mit.edu

The spread of true and false news online

Soroush Vosoughi,¹ Deb Roy,¹ Sinan Aral^{2*}

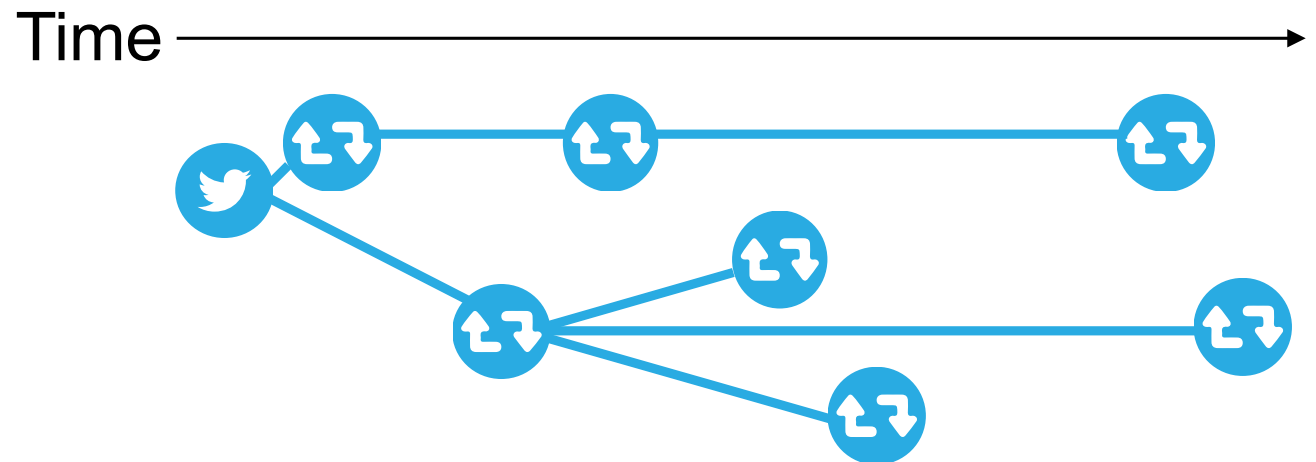
We investigated the differential diffusion of all of the verified true and false news stories distributed on Twitter from 2006 to 2017. The data comprise ~126,000 stories tweeted by ~3 million people more than 4.5 million times. We classified news as true or false using information from six independent fact-checking organizations that exhibited 95 to 98% agreement on the classifications. Falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information, and the effects were more pronounced for false political news than for false news about terrorism, natural disasters, science, urban legends, or financial information. We found that false news was more novel than true news, which suggests that people were more likely to share novel information. Whereas false stories inspired fear, disgust, and surprise in replies, true stories inspired anticipation, sadness, joy, and trust. Contrary to conventional wisdom, robots accelerated the spread of true and false news at the same rate, implying that false news spreads more than the truth because humans, not robots, are more likely to spread it.

Abstract

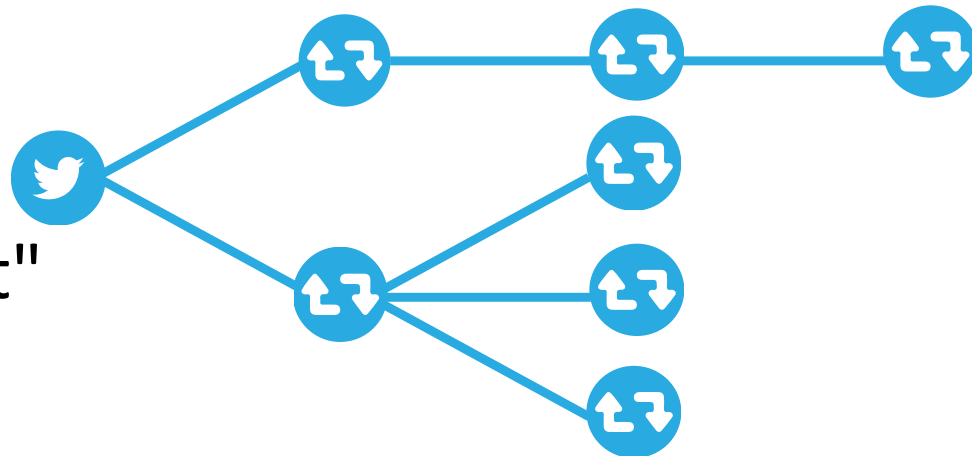
We investigated the differential diffusion of all of the verified true and false news stories distributed on Twitter from 2006 to 2017. The data comprise ~126,000 stories tweeted by ~3 million people more than 4.5 million times. We classified news as true or false using information from six independent fact-checking organizations that exhibited 95 to 98% agreement on the classifications. Falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information, and the effects were more pronounced for false political news than for false news about terrorism, natural disasters, science, urban legends, or financial information. We found that false news was more novel than true news, which suggests that people were more likely to share novel information. Whereas false stories inspired fear, disgust, and surprise in replies, true stories inspired anticipation, sadness, joy, and trust. Contrary to conventional wisdom, robots accelerated the spread of true and false news at the same rate, implying that false news spreads more than the truth because humans, not robots, are more likely to spread it.

"Rumor cascade"

- Originale



- "Discretizzata"



- → "Albero di tweet"

4 misure per quantificare diffusione

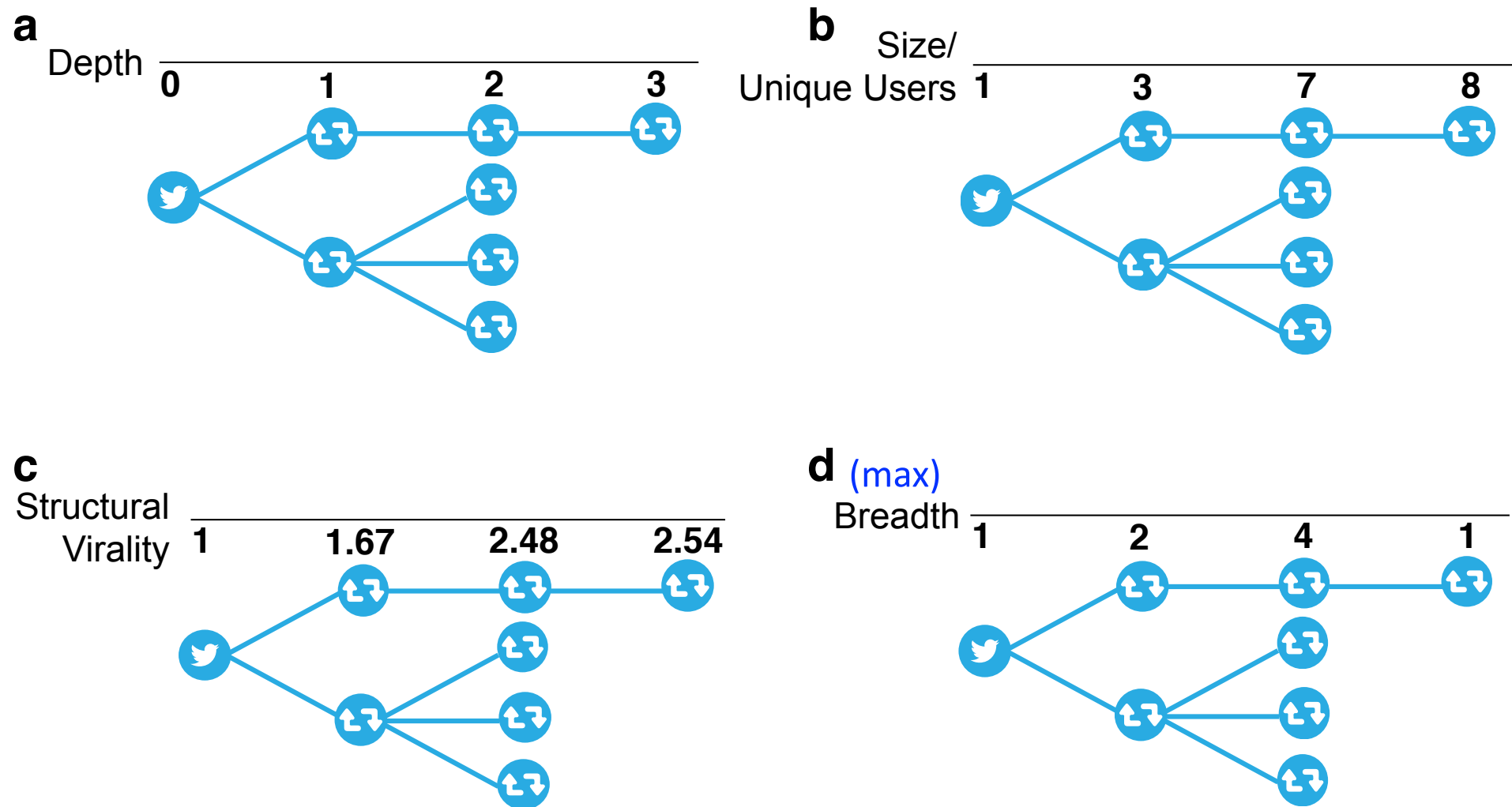
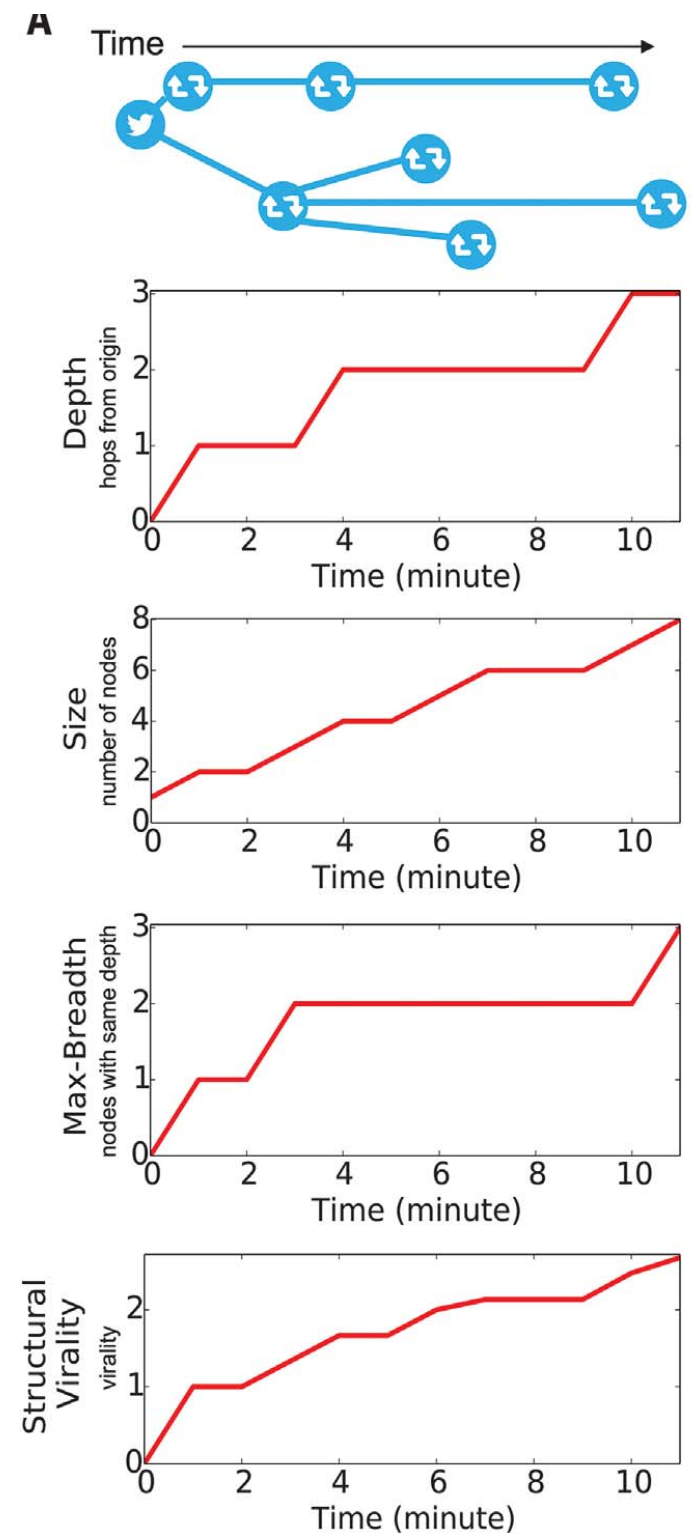


Figure S7: Depth, size, structural virality and breadth calculated for a sample cascade.

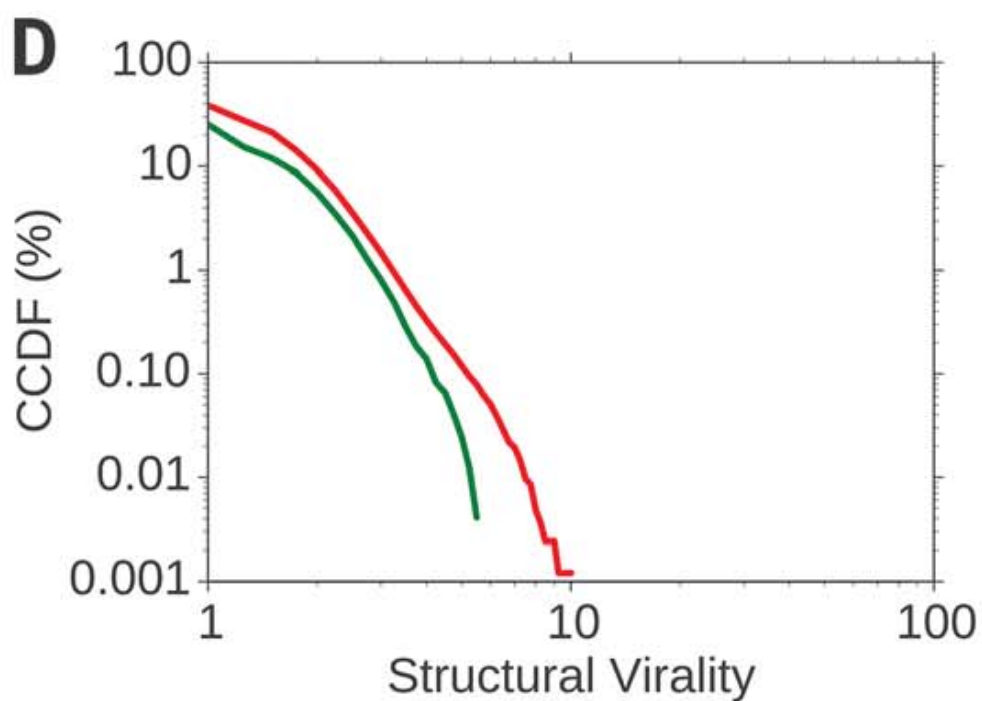
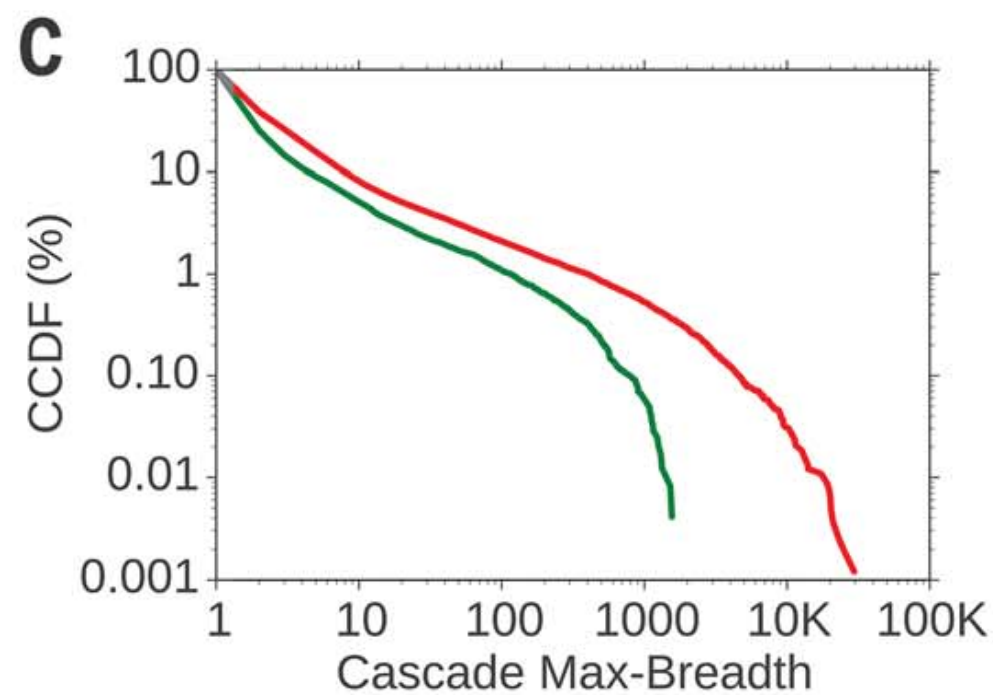
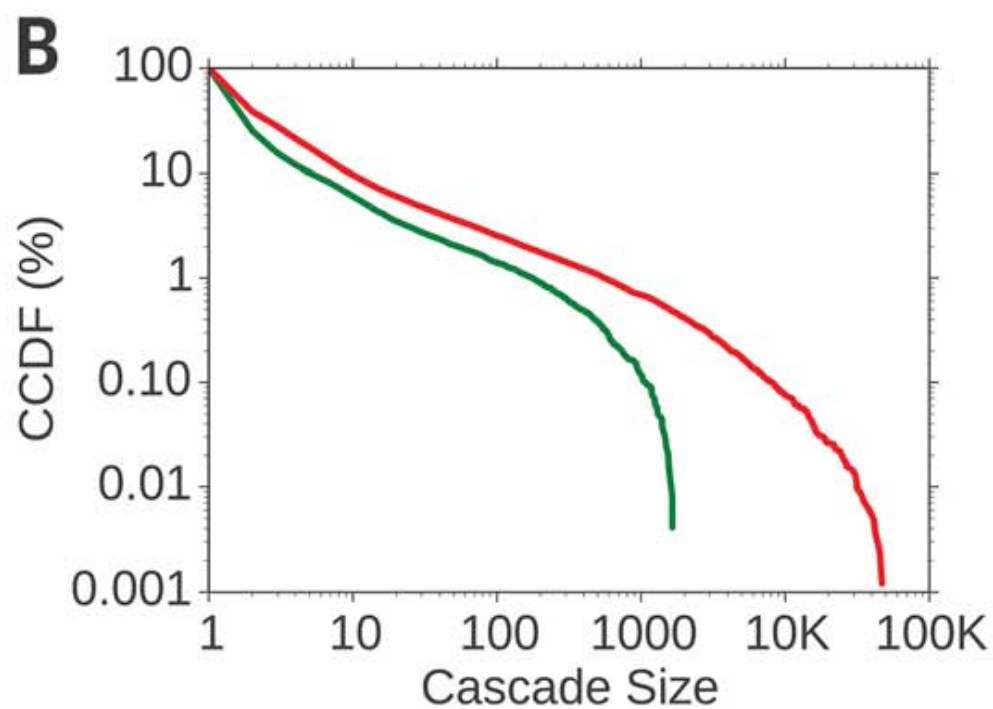
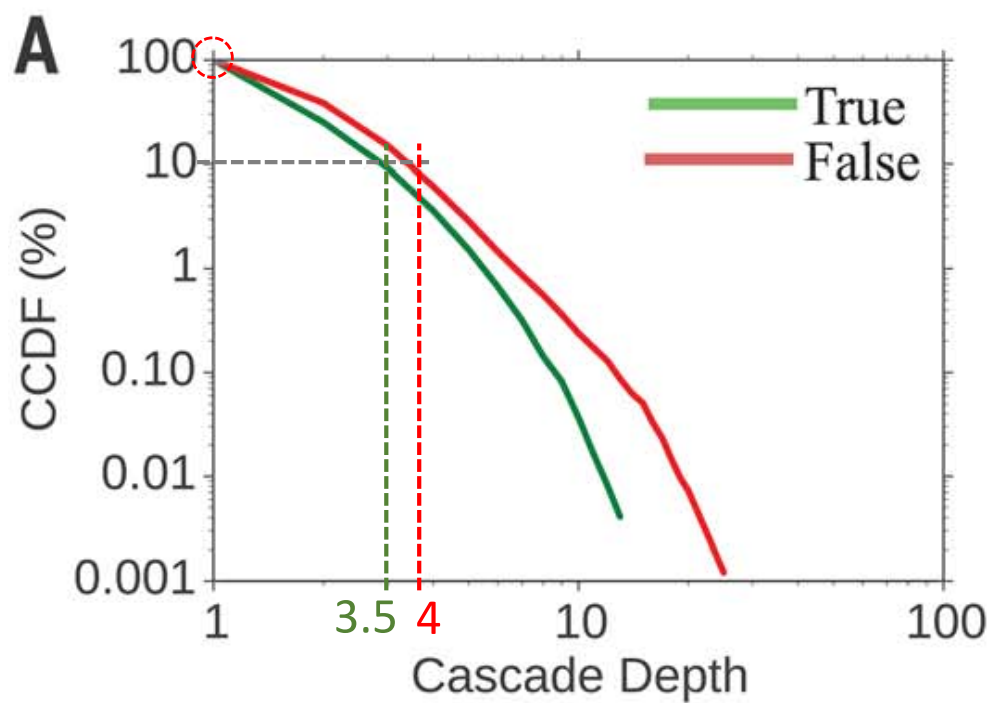
Misure: Es.

- Profondità
- Dimensione / grandezza
 - = # utenti coinvolti
- Ampiezza max.
- "Viralità strutturale"
 - (~ Quanto il rumor si è diffuso)



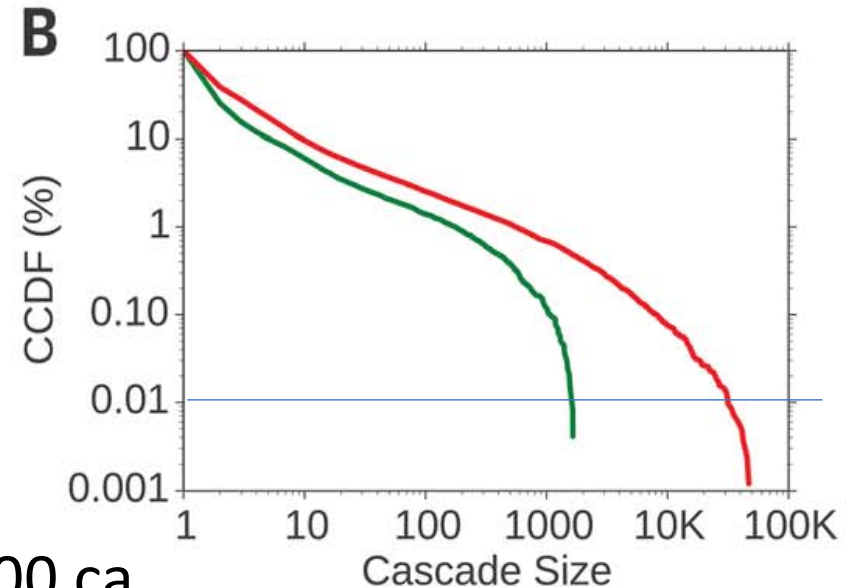
RQ

- Si diffondono di più le notizie vere o quelle false?
 - (secondo le 4 misure)
- Risultati...



Risultati

- Diffusione notizie vere e false: simili, ma
- Le curve rosse sono sempre a dx
- → notizie false hanno
 - > Depth
 - > Size
 - > Breadth
 - > Virality
- Es.:
 - Lo 0.01% delle true ha size ≥ 1000 ca
 - Lo 0.01% delle false ha size ≥ 50000 ca (!)



(altri risultati, non visti)

- News di ambito politico "peggio"
 - (rispetto a news di terrorismo, disastri naturali, scienza, leggende urbane, informazioni finanziarie)
- Notizie false più nuove, inusuali
 - (Le persone sono probabilmente più propense a diffondere informazioni nuove e sorprendenti)
- Emozioni
 - Storie false → paura, disgusto, sorpresa
 - Storie vere → attesa, tristezza, gioia
- Bot: nessuna differenza false/vere → le notizie false si diffondono di più perché le diffondono gli umani, non i bot!

Abstract (again)

We investigated the differential diffusion of all of the verified true and false news stories distributed on Twitter from 2006 to 2017. The data comprise ~126,000 stories tweeted by ~3 million people more than 4.5 million times. We classified news as true or false using information from six independent fact-checking organizations that exhibited 95 to 98% agreement on the classifications. Falsehood diffused significantly farther, faster, deeper, and more broadly than the truth in all categories of information, and the effects were more pronounced for false political news than for false news about terrorism, natural disasters, science, urban legends, or financial information. We found that false news was more novel than true news, which suggests that people were more likely to share novel information. Whereas false stories inspired fear, disgust, and surprise in replies, true stories inspired anticipation, sadness, joy, and trust. Contrary to conventional wisdom, robots accelerated the spread of true and false news at the same rate, implying that false news spreads more than the truth because humans, not robots, are more likely to spread it.

Riassunto – Ahia! 😞

- Le notizie false si diffondono di più
- E non per "colpa" degli utenti bot ma degli umani
- E i pattern di diffusione non sembrano aiutare per distinguere notizie false da vere
 - (le curve sono simili)

Secondo studio [PNAS]

- Facebook (invece di Twitter)
- 67 pagine pubbliche
 - 32 su teorie cospirazioniste (→ "false")
 - 35 su notizie scientifiche (→ "vere")
- (e 2 pagine troll)
- Download di tutti i post nel quinquennio 2010 – 2014
 - E anche tutte le "user interactions" (condivisioni, commenti, like & co. reactions)

15. Diffusione disinformazioni - S. Mizzaro

The spreading of misinformation online

Michela Del Vicario^a, Alessandro Bessi^b, Fabiana Zollo^b, Fabio Petroni^c, Antonio Scala^{a,d}, Guido Caldarelli^{a,d}, H. Eugene Stanley^e, and Walter Quattrociocchi^{a,1}

^aLaboratory of Computational Social Science, Networks Department, IMT Alti Studi Lucca, 55100 Lucca, Italy; ^bIUSS Institute for Advanced Study, 27100 Pavia, Italy; ^cSapienza University, 00185 Rome, Italy; ^dISC-CNR Uos "Sapienza," 00185 Rome, Italy; and ^eBoston University, Boston, MA 02115

Edited by Matjaz Perc, University of Maribor, Maribor, Slovenia, and accepted by the Editorial Board December 4, 2015 (received for review September 1, 2015)

The wide availability of user-provided content in online social media facilitates the aggregation of people around common interests, worldviews, and narratives. However, the World Wide Web (WWW) also allows for the rapid dissemination of unsubstantiated rumors and conspiracy theories that often elicit rapid, large, but naive social responses such as the recent case of Jade Helm 15—where a simple military exercise turned out to be perceived as the beginning of a new civil war in the United States. In this work, we address the determinants governing misinformation spreading through a thorough quantitative analysis. In particular, we focus on how Facebook users consume information related to two distinct narratives: scientific and conspiracy news. We find that, although consumers of scientific and conspiracy stories present similar consumption patterns with respect to content, cascade dynamics differ. Selective exposure to content is the primary driver of content diffusion and generates the formation of homogeneous clusters, i.e., "echo chambers." Indeed, homogeneity appears to be the primary driver for the diffusion of contents and each echo chamber has its own cascade dynamics. Finally, we introduce a data-driven percolation model mimicking rumor spreading and we show that homogeneity and polarization are the main determinants for predicting cascades' size.

misinformation | virality | Facebook | rumor spreading | cascades

The massive diffusion of sociotechnical systems and micro-blogging platforms on the World Wide Web (WWW) creates a direct path from producers to consumers of content, i.e., allows disintermediation, and changes the way users become informed, debate, and form their opinions (1–5). This disintermediated environment can foster confusion about causation, and thus encourage speculation, rumors, and mistrust (6). In 2011 a blogger claimed that global warming was a fraud designed to diminish liberty and weaken democracy (7). Misinformation about the Ebola epidemic has caused confusion among healthcare workers (8). Jade Helm 15, a simple military exercise, was perceived on the Internet as the beginning of a new civil war in the United States (9).

Recent works (10–12) have shown that increasing the exposure of users to unsubstantiated rumors increases their tendency to be credulous.

According to ref. 13, beliefs formation and revision is influenced by the way communities attempt to make sense of events or facts. Such a phenomenon is particularly evident on the WWW where users, embedded in homogeneous clusters (14–16), process information through a shared system of meaning (10, 11, 17, 18) and trigger collective framing of narratives that are often biased toward self-confirmation.

In this work, through a thorough quantitative analysis on a massive dataset, we study the determinants behind misinformation diffusion. In particular, we analyze the cascade dynamics of Facebook users when the content is related to very distinct narratives: conspiracy theories and scientific information. On the one hand, conspiracy theories simplify causation, reduce the complexity of reality, and are formulated in a way that is able to tolerate a certain level of uncertainty (19–21). On the other hand, scientific information disseminates scientific advances and exhibits the process of scientific thinking. Notice that we do not focus on the quality of the information but rather on the possibility of verification. Indeed,

the main difference between the two is content verifiability. The generators of scientific information and their data, methods, and outcomes are readily identifiable and available. The origins of conspiracy theories are often unknown and their content is strongly disengaged from mainstream society and sharply divergent from recommended practices (22), e.g., the belief that vaccines cause autism.

Massive digital misinformation is becoming pervasive in online social media to the extent that it has been listed by the World Economic Forum (WEF) as one of the main threats to our society (23). To counteract this trend, algorithmic-driven solutions have been proposed (24–29), e.g., Google (30) is developing a trustworthiness score to rank the results of queries. Similarly, Facebook has proposed a community-driven approach where users can flag false content to correct the newsfeed algorithm. This issue is controversial, however, because it raises fears that the free circulation of content may be threatened and that the proposed algorithms may not be accurate or effective (10, 11, 31). Often conspiracists will denounce attempts to debunk false information as acts of misinformation.

Whether a claim (either substantiated or not) is accepted by an individual is strongly influenced by social norms and by the claim's coherence with the individual's belief system—i.e., confirmation bias (32, 33). Many mechanisms animate the flow of false information that generates false beliefs in an individual, which, once adopted, are rarely corrected (34–37).

In this work we provide important insights toward the understanding of cascade dynamics in online social media and in particular about misinformation spreading.

We show that content-selective exposure is the primary driver of content diffusion and generates the formation of homogeneous

Significance

The wide availability of user-provided content in online social media facilitates the aggregation of people around common interests, worldviews, and narratives. However, the World Wide Web is a fruitful environment for the massive diffusion of unverified rumors. In this work, using a massive quantitative analysis of Facebook, we show that information related to distinct narratives—conspiracy theories and scientific news—generates homogeneous and polarized communities (i.e., echo chambers) having similar information consumption patterns. Then, we derive a data-driven percolation model of rumor spreading that demonstrates that homogeneity and polarization are the main determinants for predicting cascades' size.

Author contributions: M.D.V., A.B., F.Z., A.S., G.C., H.E.S., and W.Q. designed research; M.D.V., A.B., F.Z., H.E.S., and W.Q. performed research; M.D.V., A.B., F.Z., F.P., and W.Q. contributed new reagents/analytic tools; M.D.V., A.B., F.Z., A.S., G.C., H.E.S., and W.Q. analyzed data; and M.D.V., A.B., F.Z., A.S., G.C., H.E.S., and W.Q. wrote the paper.

The authors declare no conflict of interest.

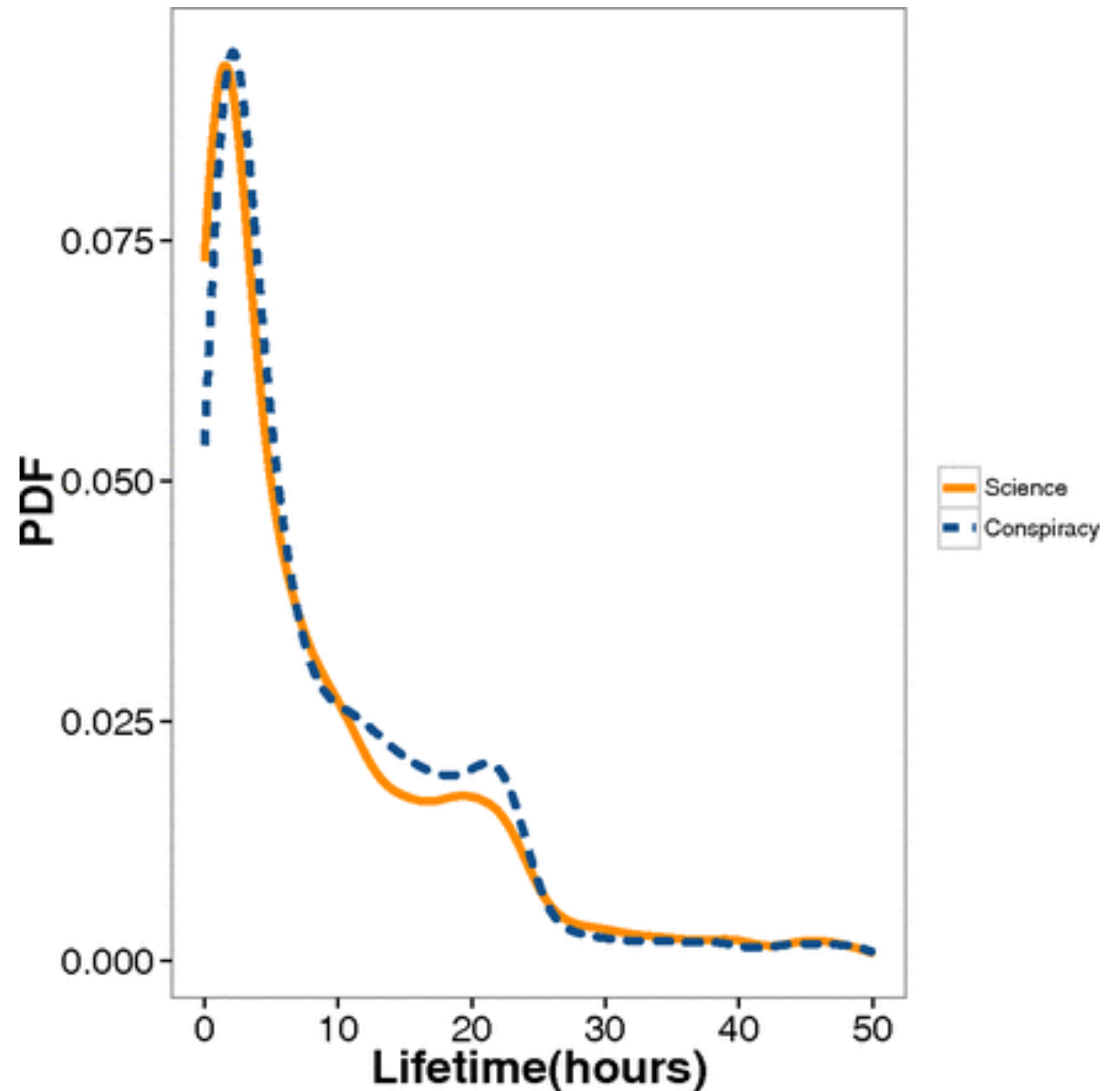
This article is a PNAS Direct Submission. M.P. is a guest editor invited by the Editorial Board.

Freely available online through the PNAS open access option.

¹To whom correspondence should be addressed. Email: walterquattrociocchi@gmail.com. This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1517441113/-DCSupplemental.

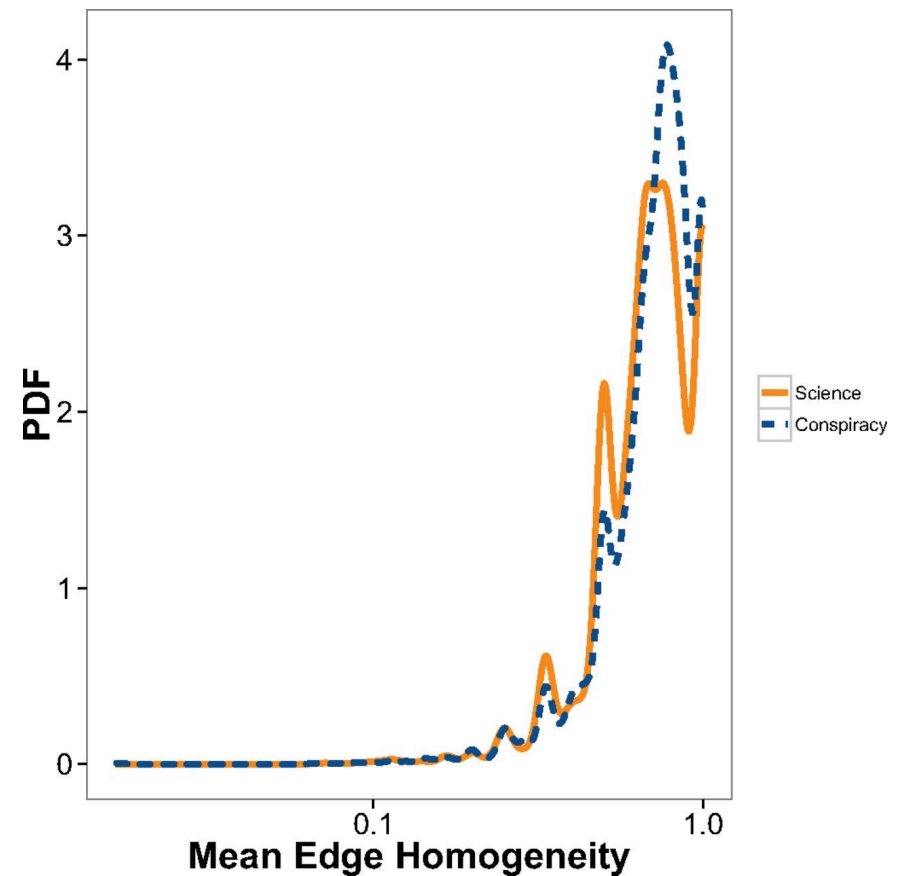
Durata

- Tempo fra prima e ultima condivisione
- Pattern di condivisione temporale: simile



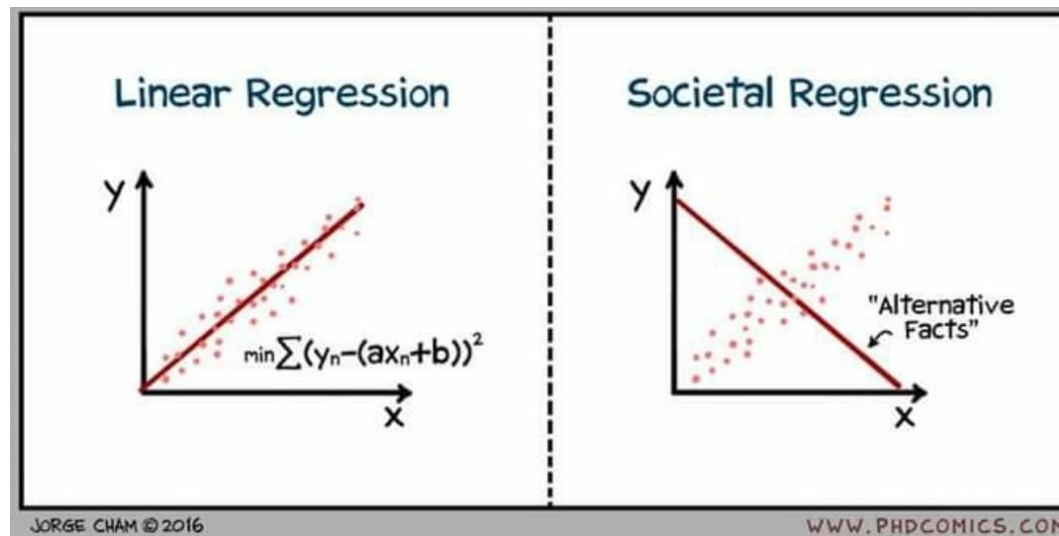
Due comunità

- La maggior parte dei post ha **alta "omogeneità"**
 - Nodo A condivide post di nodo B
 - Def. 2 Comunità: conspiracy / science (chi mette like a post cospirazionisti / scientifici)
 - Omogeneità: misura di quanto i due nodi sono nella stessa comunità
 - Omogeneità $\sim 1 \rightarrow$ i due nodi o sono entrambi nella comunità conspiracy o entrambi nella comunità science
 - Condivido quasi solo post di chi la pensa come me
- Si formano due comunità intorno ad argomenti
 - Cospirazionisti
 - Scientifici
- Separate, segregate



"Echo chambers"

- Camere dell'eco, Camera di risonanza
- Sorta di evidenza sperimentale delle Filter bubble, Bolle informative
 - Eli Pariser (2011)
 - <https://www.youtube.com/watch?v=B8ofWfx525s>
- Alta omogeneità: i contenuti tendono a circolare solo all'interno della rispettiva comunità / camera dell'eco



Terzo studio [PLOS ONE]



- Ma se si prova a spiegare a chi diffonde notizie false che sta sbagliando, magari smette...
- Attività di Debunking



OPEN ACCESS

Citation: Zollo F, Bessi A, Del Vicario M, Scala A, Caldarelli G, Shekhtman L, et al. (2017) Debunking in a world of tribes. PLoS ONE 12(7): e0181821. <https://doi.org/10.1371/journal.pone.0181821>

Editor: Jose Javier Ramasco, Instituto de Física Interdisciplinar y Sistemas Complejos, SPAIN

Received: February 21, 2016

Accepted: May 13, 2017

Published: July 24, 2017

Copyright: © 2017 Zollo et al. This is an open access article distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Data Availability Statement: The entire data collection process has been carried out exclusively by means of the Facebook Graph API, which are publicly available. Further details about data collection are provided in the Methods section of the paper, together with the complete list of pages.

Funding: Funding for this work was provided by EU FET project MULTIPLEX nr. 317532, SIMPOL nr. 610704, DOLFINS nr. 640772, SOBIGDATA 654024, IMT/xtrapola Srl (P0082). SH and LS were supported by the Israel Ministry of Science and Technology, the Japan Science and Technology Agency, the Italian Ministry of Foreign

RESEARCH ARTICLE

Debunking in a world of tribes

Fabiana Zollo^{1,2*}, Alessandro Bessi³, Michela Del Vicario², Antonio Scala^{2,4}, Guido Caldarelli², Louis Shekhtman⁵, Shlomo Havlin⁵, Walter Quattrociocchi²

1 Ca' Foscari University of Venice, Venezia, Italy, **2** IMT School for Advanced Studies, Lucca, Italy, **3** IUSS, Pavia, Italy, **4** ISC-CNR, Roma, Italy, **5** Bar-Ilan University, Ramat Gan, Israel

* fabiana.zollo@unive.it

Abstract

Social media aggregate people around common interests eliciting collective framing of narratives and worldviews. However, in such a disintermediated environment misinformation is pervasive and attempts to *debunk* are often undertaken to contrast this trend. In this work, we examine the effectiveness of debunking on Facebook through a quantitative analysis of 54 million users over a time span of five years (Jan 2010, Dec 2014). In particular, we compare how users usually consuming proven (scientific) and unsubstantiated (conspiracy-like) information on Facebook US interact with specific debunking posts. Our findings confirm the existence of echo chambers where users interact primarily with either conspiracy-like or scientific pages. However, both groups interact similarly with the information within their echo chamber. Then, we measure how users from both echo chambers interacted with 50,220 debunking posts accounting for both users consumption patterns and the sentiment expressed in their comments. Sentiment analysis reveals a dominant negativity in the comments to debunking posts. Furthermore, such posts remain mainly confined to the scientific echo chamber. Only few conspiracy users engage with corrections and their liking and commenting rates on conspiracy posts increases after the interaction.

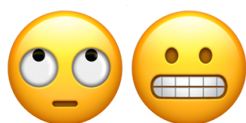
Introduction

Socio-technical systems and microblogging platforms such as Facebook and Twitter have created a direct path from producers to consumers of content, changing the way users get informed, debate ideas, and shape their worldviews [1–6]. Misinformation on online social media is pervasive and represents one of the main threats to our society according to the World Economic Forum [7, 8]. The diffusion of false rumors affects public perception of reality as well as the political debate [9]. Indeed, links between vaccines and autism, the belief that 9/11 was an inside job, or the more recent case of Jade Helm 15—a simple military exercise that was perceived as the imminent threat of the civil war in the US—are just few examples of the consistent body of the collective narratives grounded on unsubstantiated information.

Confirmation bias plays a pivotal role in cascades dynamics and facilitates the emergence of echo chambers [10]. Indeed, users online show the tendency a) to select information that adheres to their system of beliefs even when containing parodistic jokes; and b) to join polarized groups [11]. Recently, researches have shown [12–17] that continued exposure to

Purtroppo...

- Confirmation bias, Pregiudizio di conferma
- Si crede solo a quello che si sa già
 - A ciò che conferma, è compatibile con, quello che si sa già
- Informazioni dissenzienti, in contrasto, sono ignorate
- Debunking sembra inutile
 - Quando non controproducente
 - *«Mai discutere con un'idiota. Ti trascina al suo livello e ti batte con l'esperienza» (Oscar Wilde)*



*Mai discutere con un idiota.
Ti trascina al suo livello
e ti batte con l'esperienza.
(Oscar Wilde)*



www.giovannilucarelli.it

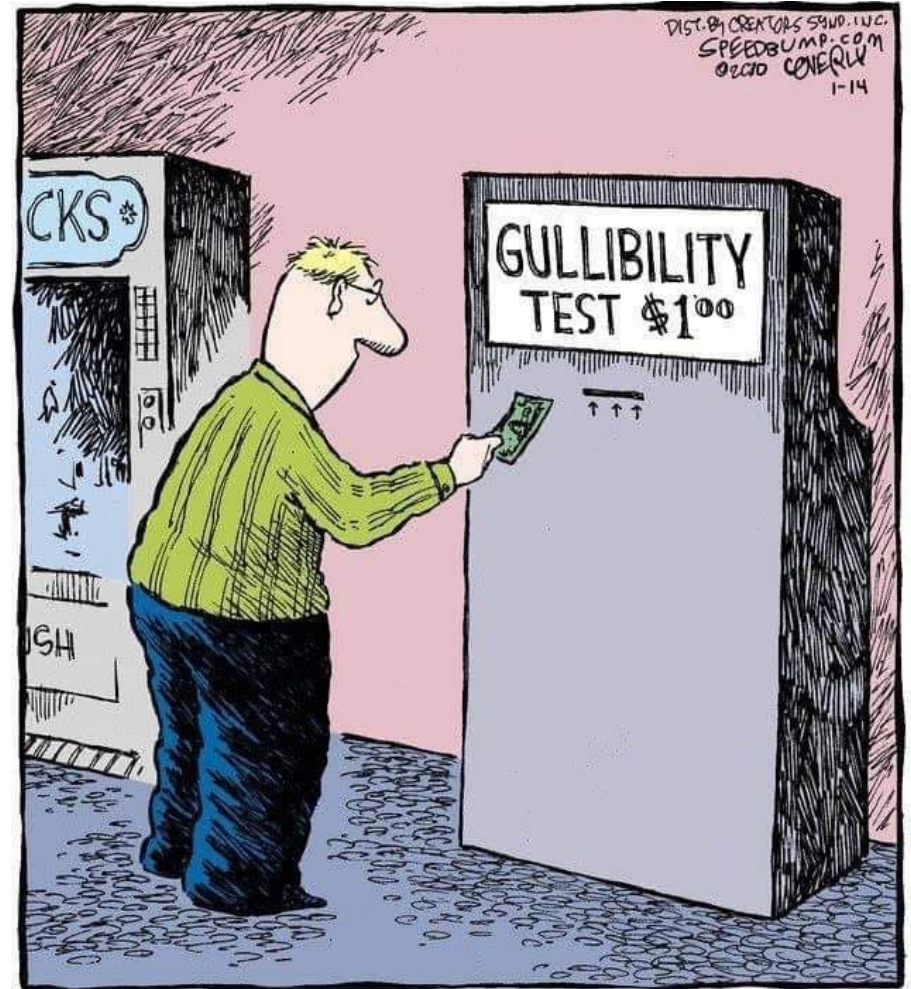


BUFALA

OSCAR WILDE NON HA MAI DETTO QUESTA FRASE, CHE VIENE DA UN BRANO DELLA BIBBIA, NEGLI ANNI ATTRIBUITA A SVARIATI AUTORI, MA PUR ERRATA LA CITAZIONE IL CONCETTO RESTA VALIDO

Scaletta

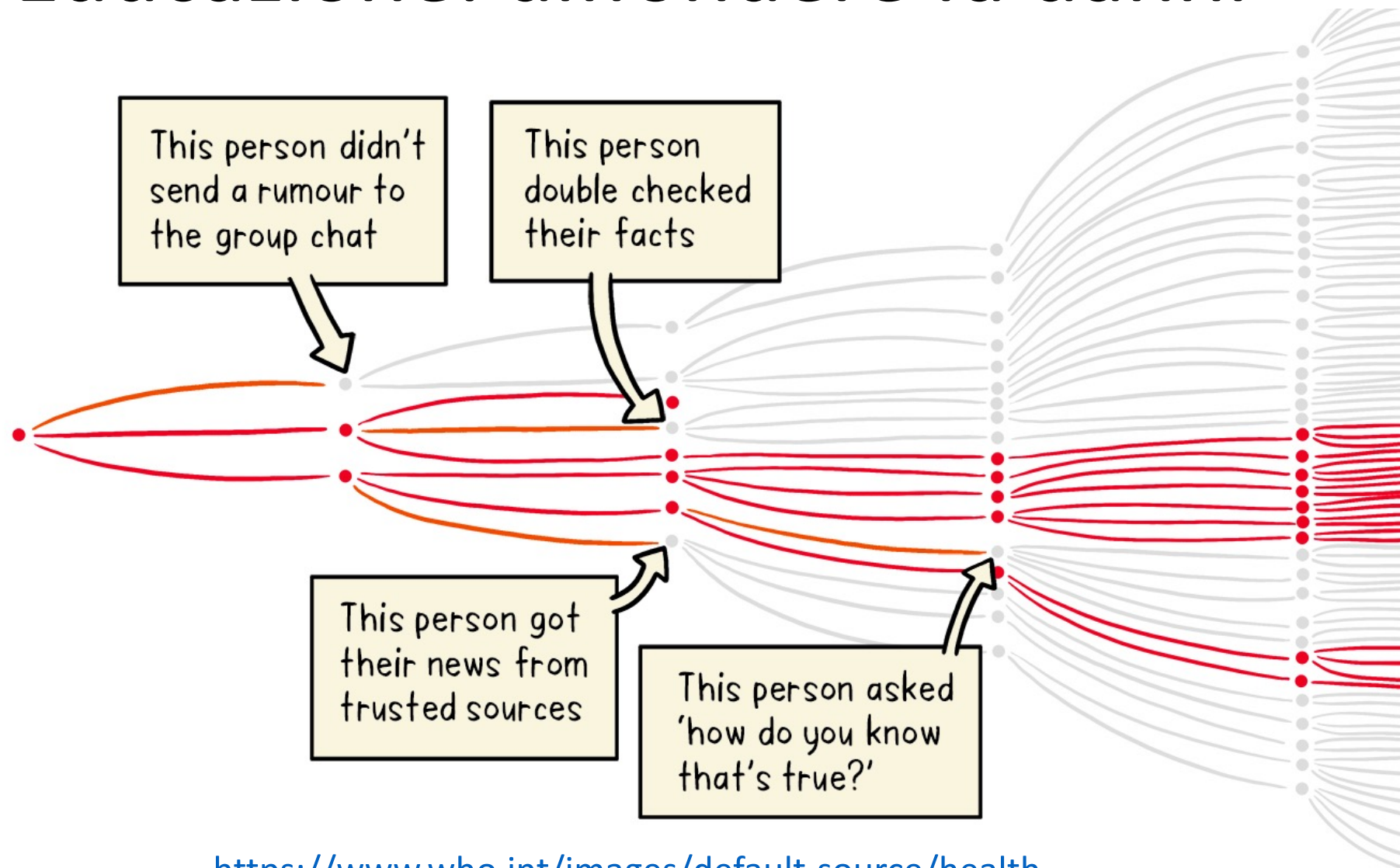
- Intro
 - Terminologia/Definizioni
 - Motivazioni
- Modelli epidemici
- 3 studi sperimentali



Riassunto

- La disinformazione è un problema importante
- E... senza speranza di soluzione?
- O almeno di difficile soluzione ☹
 - Le notizie false si diffondono di più di quelle vere
 - E non è neppure colpa dei bot
 - Si formano camere dell'eco in cui uno sente solo quello che vuol sentire / che sa già / che dice (confirmation bias)
 - Debunking è inutile
- Serve probabilmente uno sforzo importante e congiunto
 - Tecnologie (ML, classificazione)
 - Leggi
 - Educazione, ...
- Ci si sta comunque lavorando (ne riparleremo...)

Educazione: diffondere fa danni



<https://www.who.int/images/default-source/health-topics/coronavirus/spotlight-misinformation.tmb-1920v.png>